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Shimosaka et al.

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(54) **MEDIUM CONVEYANCE DEVICE**

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(73) Assignee: **PFU LIMITED**, Kahoku (JP)

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B65H 3/06 (2006.01)
(Continued)

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CPC . ***B65H 3/0638*** (2013.01); *B65H 2404/13161* (2013.01); *B65H 2404/13163* (2013.01); *B65H 2801/12* (2013.01)

(58) **Field of Classification Search**
CPC B65H 3/68; B65H 3/66; B65H 3/0638; B65H 3/5276; B65H 3/5284; B65H 3/56;
(Continued)

(56) **References Cited**

U.S. PATENT DOCUMENTS

2006/0097441 A1* 5/2006 Shikan B65H 1/24 271/118
2008/0203652 A1* 8/2008 Yasukawa B65H 3/063 271/117
(Continued)

FOREIGN PATENT DOCUMENTS

JP H02-305741 A 12/1990
JP 06179538 A * 6/1994
(Continued)

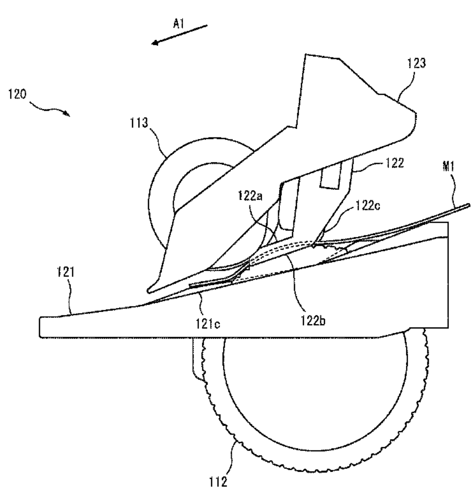
OTHER PUBLICATIONS

International Search Report dated Aug. 3, 2021 for corresponding PCT Application No. PCT/JP2021/022726 (3 pages) with English Translation (2 pages).
(Continued)

Primary Examiner — Luis A Gonzalez
(74) *Attorney, Agent, or Firm* — WOMBLE BOND DICKINSON (US) LLP

(57) **ABSTRACT**

Provided is a medium conveying apparatus to enable suitable feeding of a plurality of types of media. A medium conveying apparatus includes a guide member including an opening or a notch and configured to form a conveyance surface of a medium, a feed roller located inside the opening or the notch and configured to feed the medium, a separation roller located to face the feed roller, and protruding parts located on left and right sides of the separation roller relative to a medium conveying direction. The guide member includes recessed parts positioned on left and right sides of the feed roller relative to the medium conveying direction. The protruding parts are located at positions facing the opening or the notch, or the recessed parts in such a way that the medium is conveyed between the protruding parts and the recessed parts. A tip of each of the protruding parts is
(Continued)



positioned on the feed roller side of a nip part of the feed roller and the separation roller.

11 Claims, 25 Drawing Sheets

(51) **Int. Cl.**
B65H 3/54 (2006.01)
B65H 3/66 (2006.01)

(58) **Field of Classification Search**
CPC ... B65H 3/565; B65H 3/54; B65H 2301/5122
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2013/0328261	A1*	12/2013	Cheng	B65H 3/56 271/18
2014/0131939	A1*	5/2014	Shih	B65H 3/0653 271/126
2014/0203494	A1*	7/2014	Okumura	B65H 3/56 271/18
2018/0179006	A1	6/2018	Kogi et al.	
2020/0071100	A1*	3/2020	Namiki	B65H 7/02
2021/0099592	A1*	4/2021	Nishimura	H04N 1/00665
2021/0155439	A1*	5/2021	Namiki	B65H 7/125

FOREIGN PATENT DOCUMENTS

JP	H06-179540	A	6/1994
JP	2007-076875	A	3/2007
JP	2016-179872	A	10/2016
JP	2018-108871	A	7/2018

OTHER PUBLICATIONS

Written Opinion dated Aug. 3, 2021 for corresponding PCT Application No. PCT/JP2021/022726 (3 pages) with English Translation (3 pages).
Preliminary Report on Patentability dated Dec. 14, 2023 for corresponding PCT Application No. PCT/JP2021/022726 (4 pages) with English Translation (4 pages).
Japanese Office Action dated Feb. 18, 2025 regarding Japanese Patent Application No. 2024-096177 corresponding to U.S. Appl. No. 18/568,746 (3 pages) with English Translation (6 pages).
Japanese Office Action dated Jul. 8, 2025 regarding Japanese Patent Application No. 2024-096177 corresponding to U.S. Appl. No. 18/568,746 (3 pages) with English Translation (5 pages).

* cited by examiner

FIG. 1

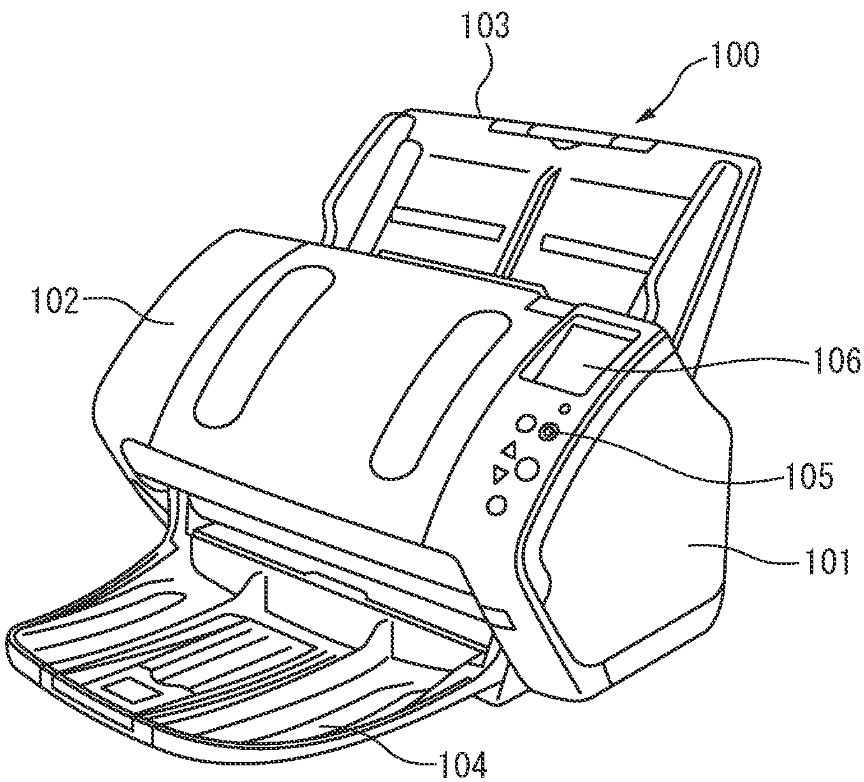


FIG. 2

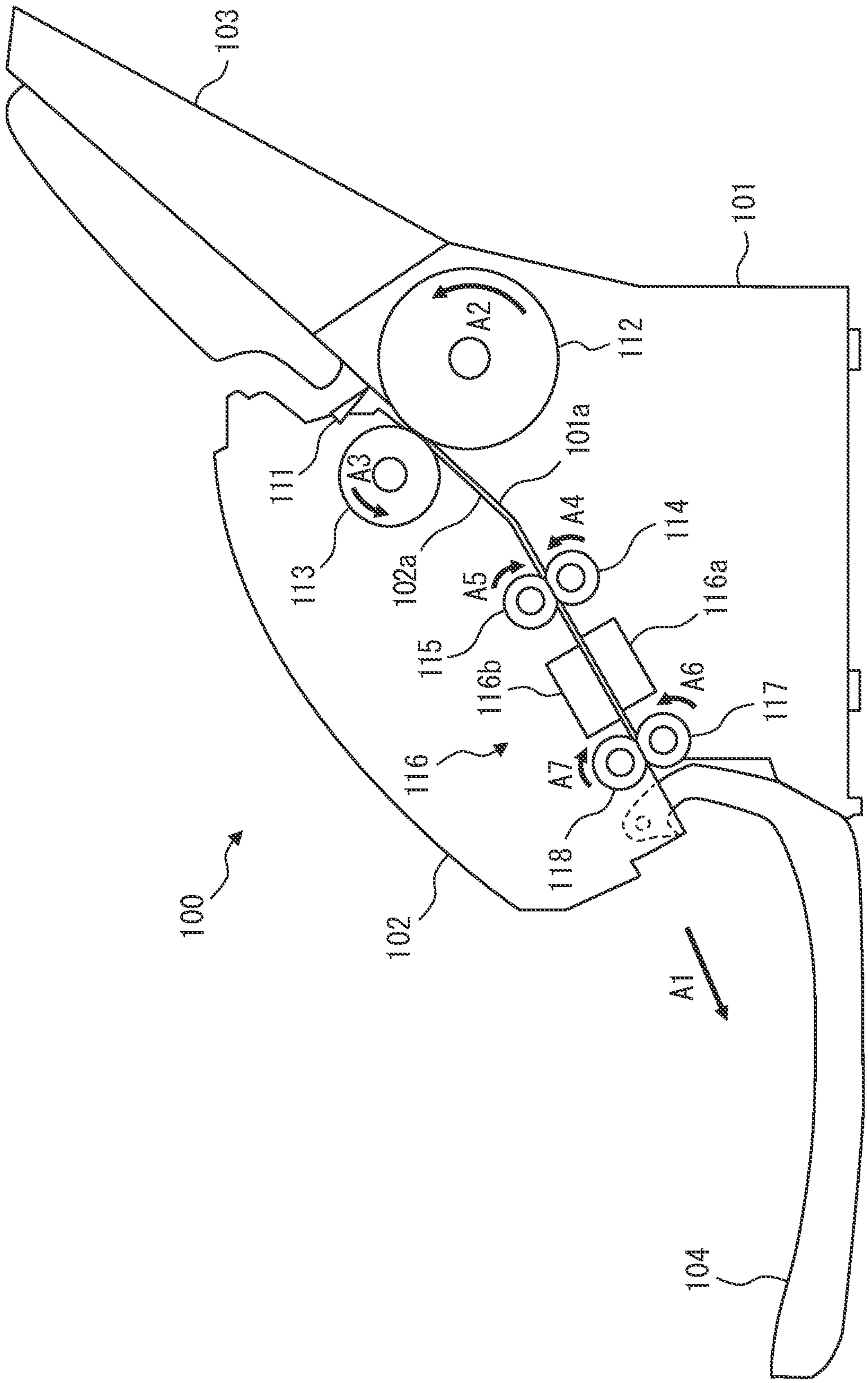

100
101
101a
102
102a
103
104
111
112
113
114
115
116
116a
116b
117
118
A1
A2
A3
A4
A5
A6
A7

FIG. 3

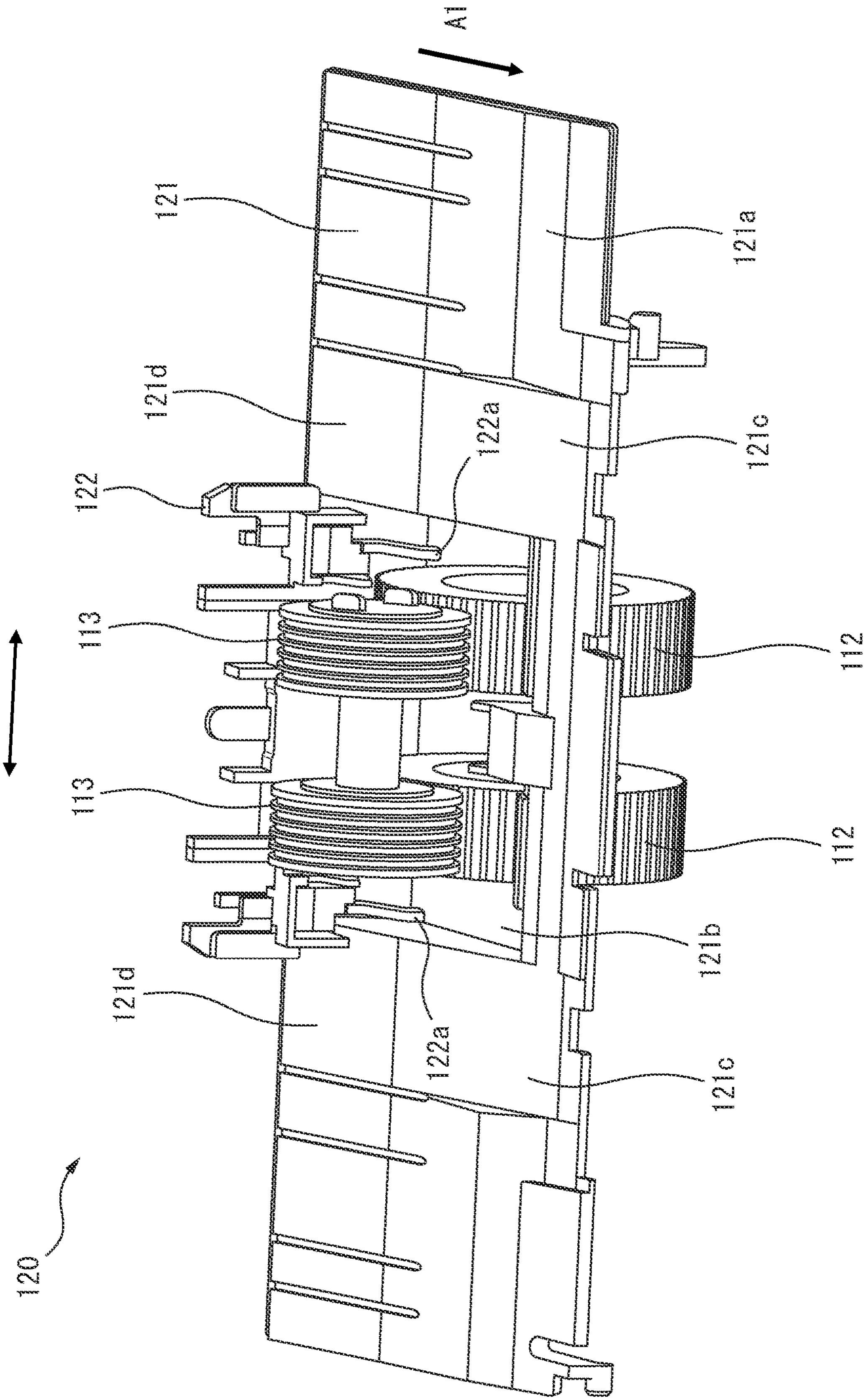

A1
121
121a
121d
122a
121c
122
113
A8
112
113
112
121b
121d
122a
121c
120

FIG. 4

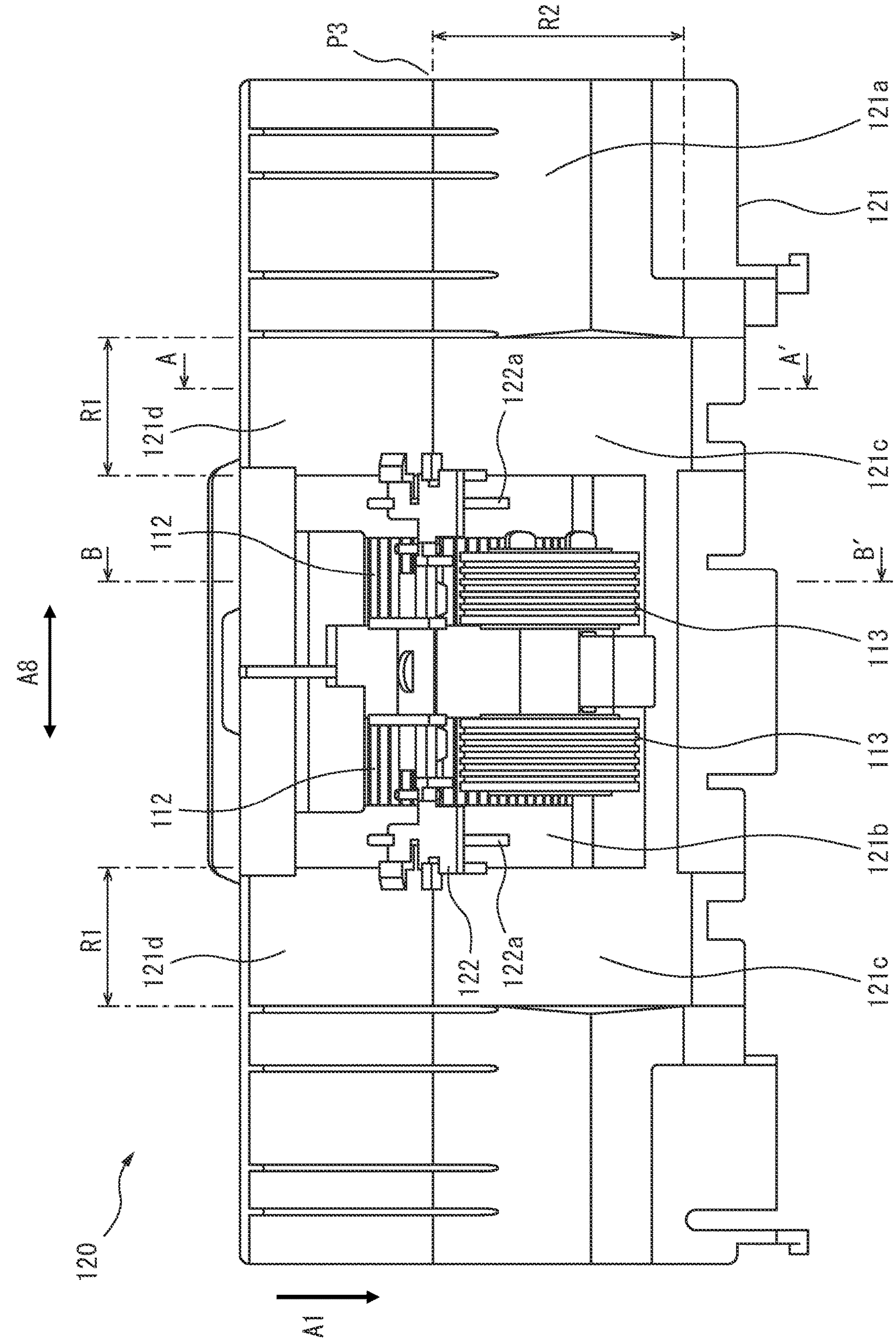
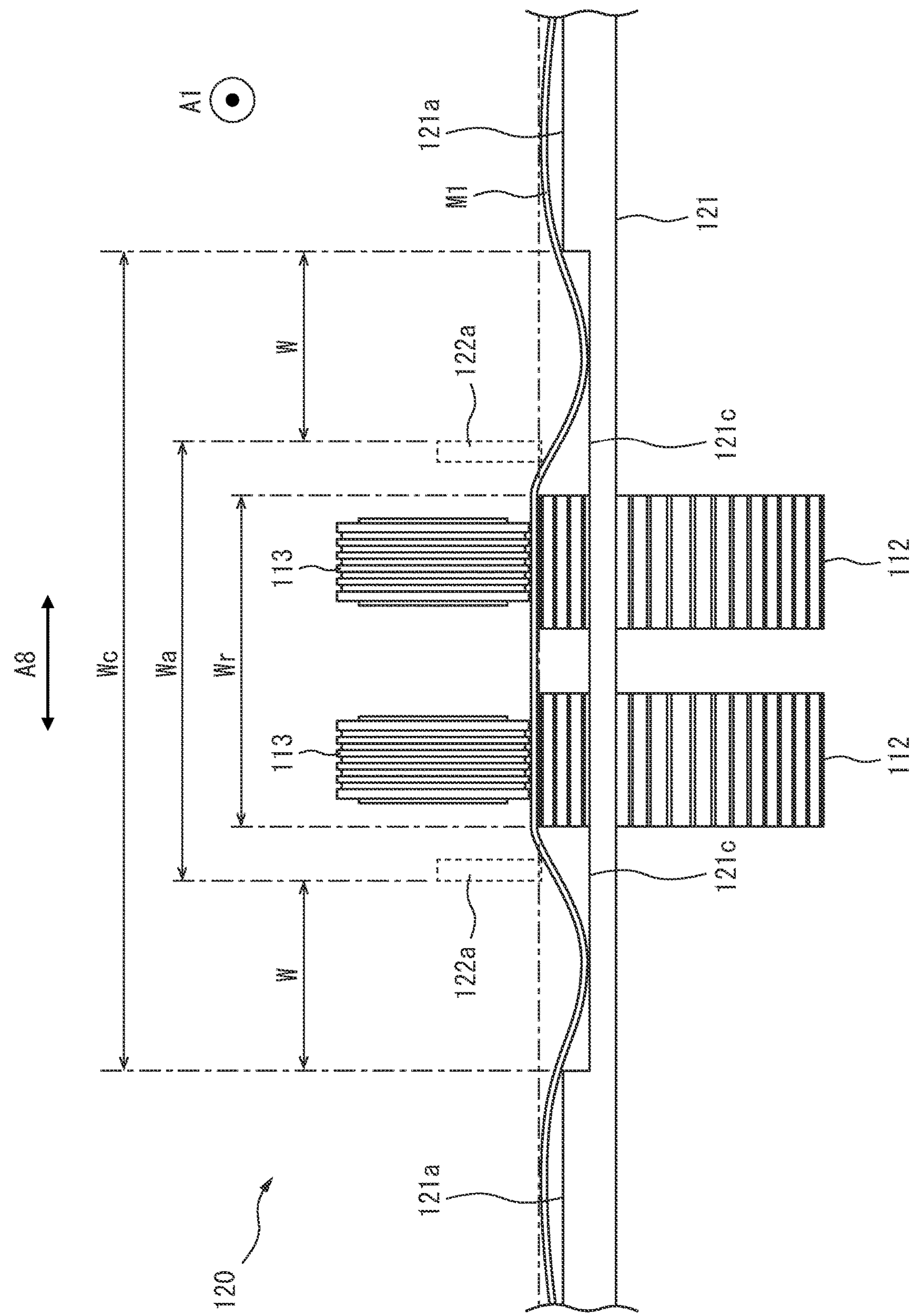

P3
R2
121a
121
R1
A
A'
122a
121d
121c
112
B
B'
A8
113
113
112
121b
R1
121d
122
122a
121c
120
A1

FIG. 5



A8
A1
Wc
Wa
Wr
W
W
113
113
112
112
122a
122a
121c
121c
121a
121a
M1
121
120

FIG. 6

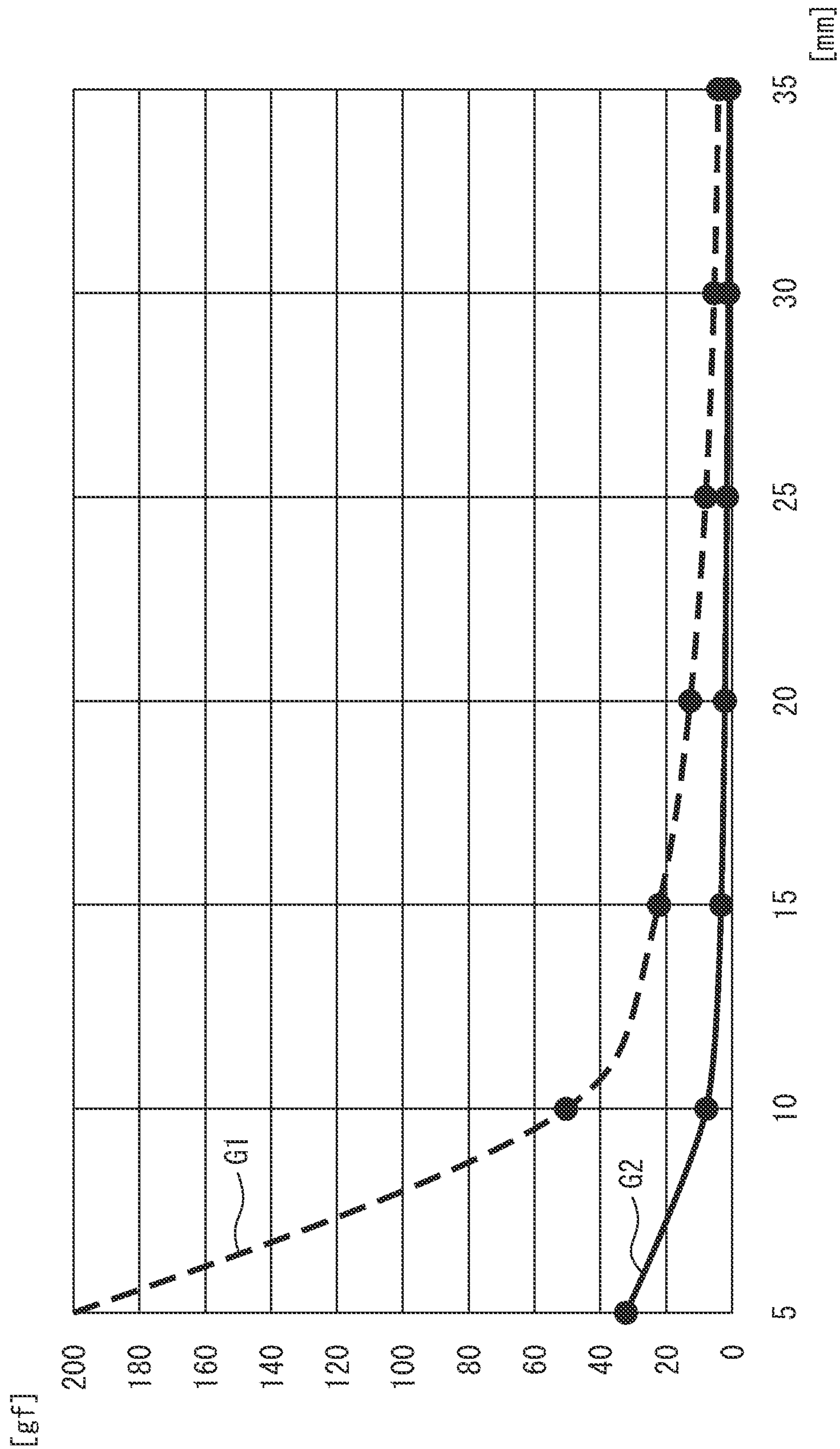


FIG. 7

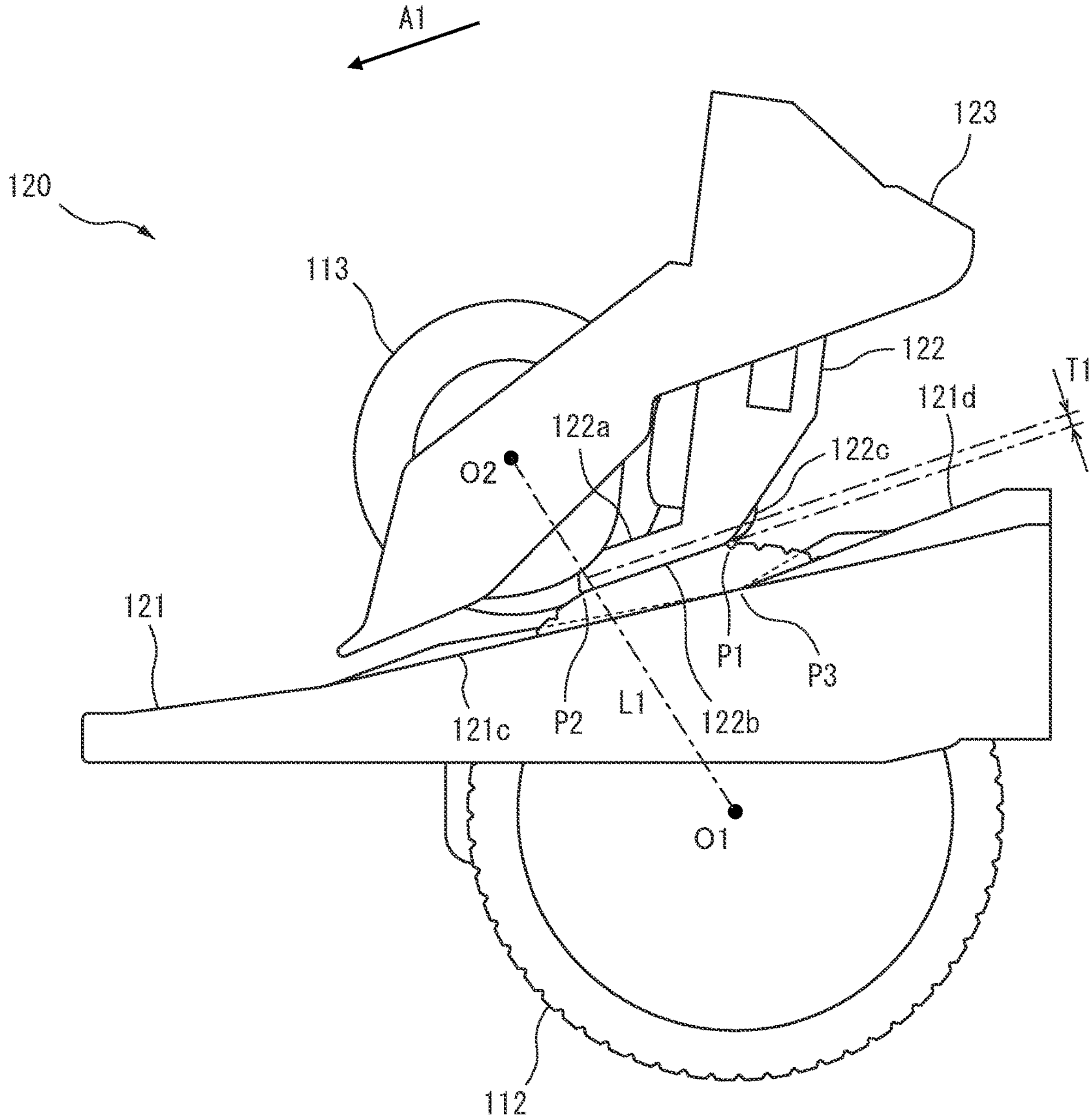
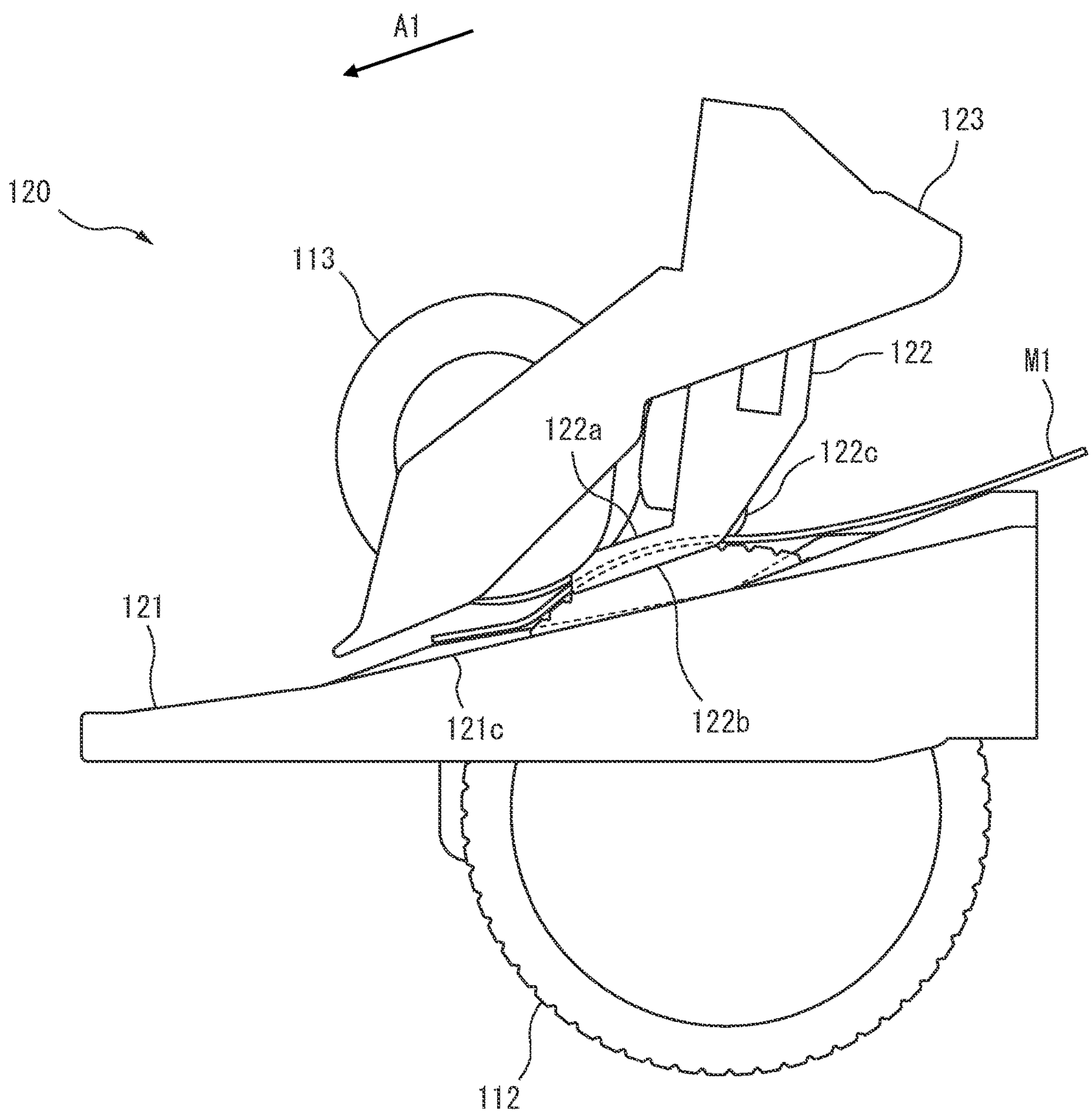

A1
123
120
113
122
T1
121d
122a
122c
O2
121
P1
P3
121c
P2
L1
122b
O1
112

FIG. 8



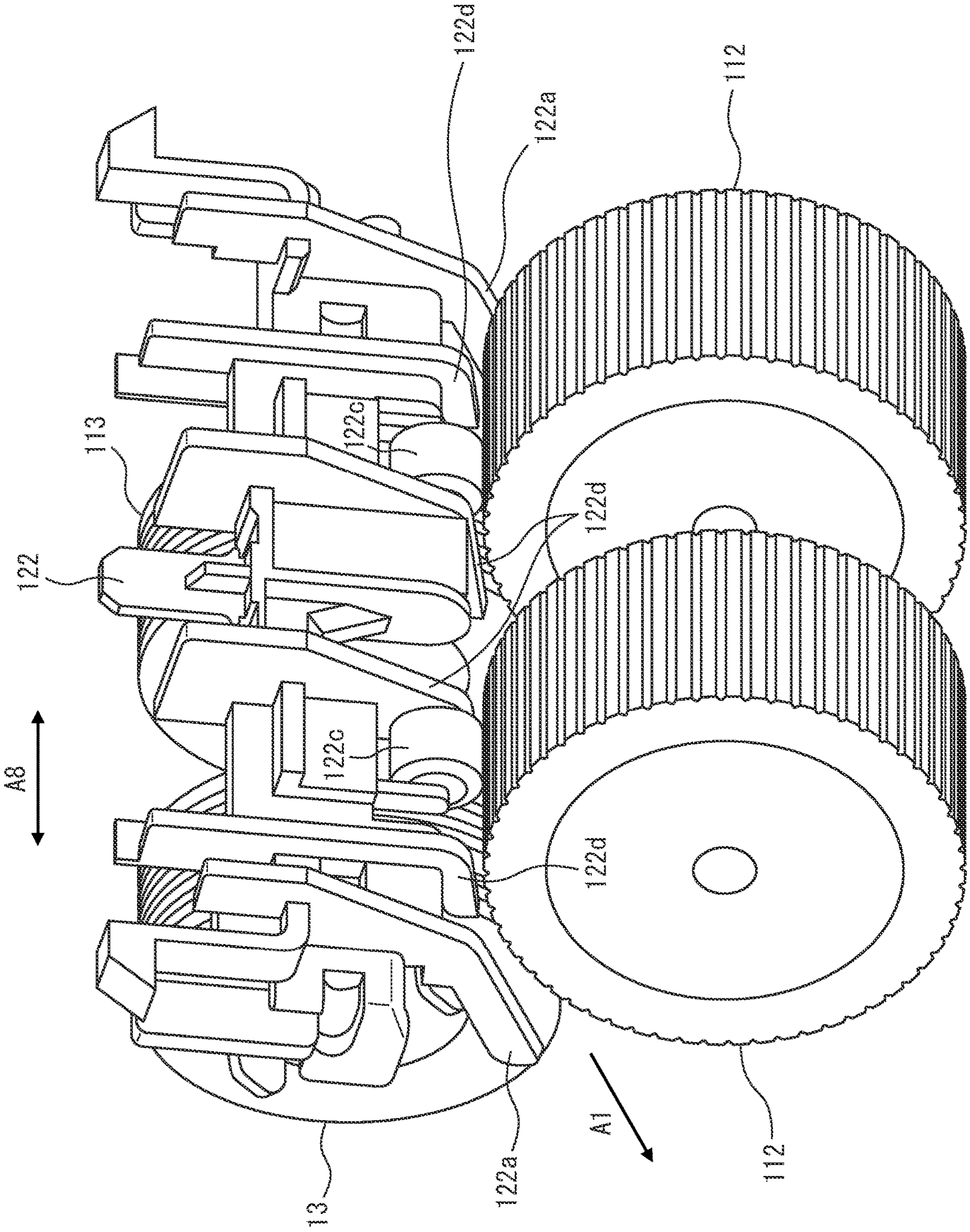

122d
112
122a
113
122c
122d
122
122d
A8
122c
122d
122a
A1
112
113

FIG. 9

FIG. 10

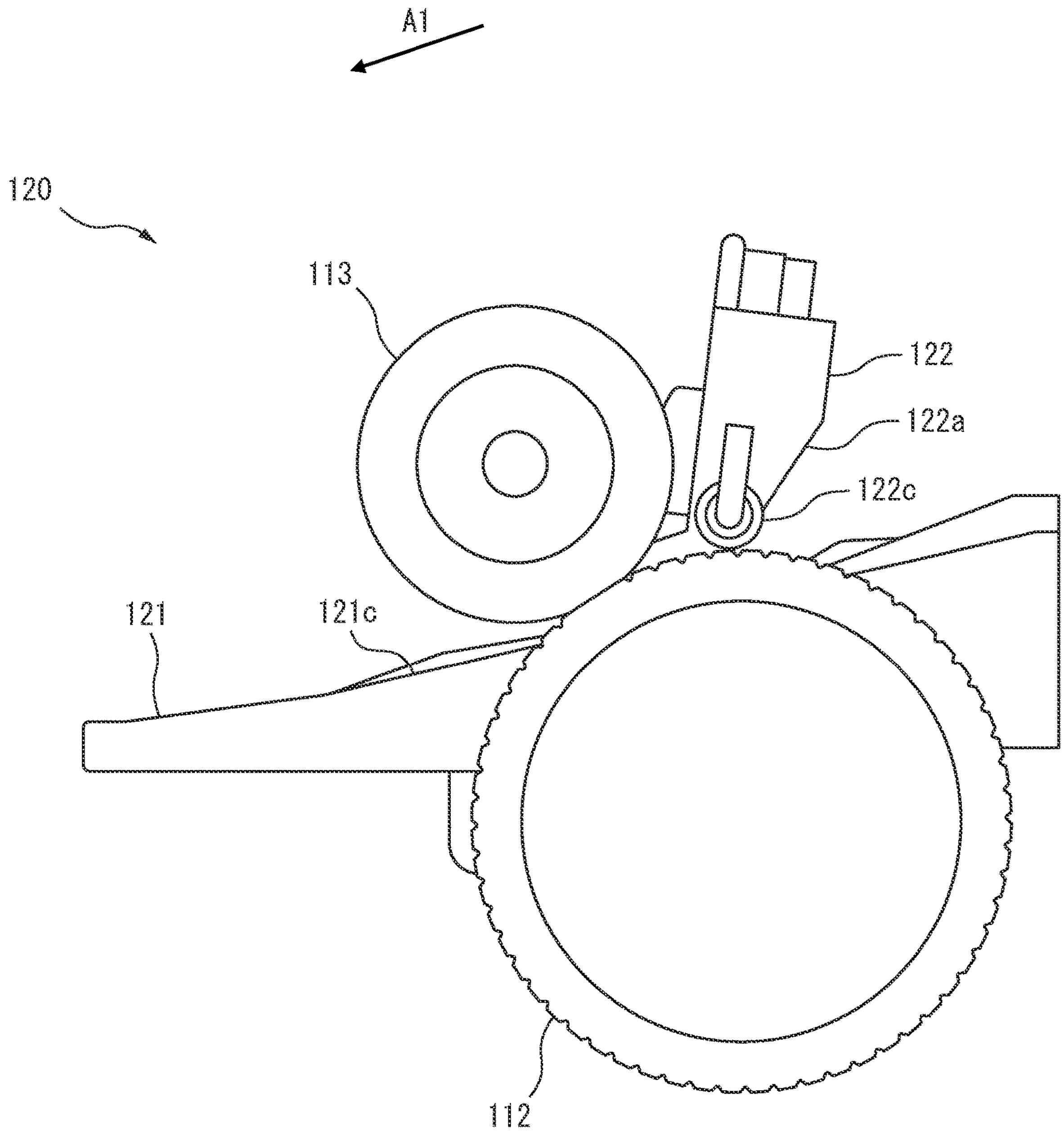


FIG. 11

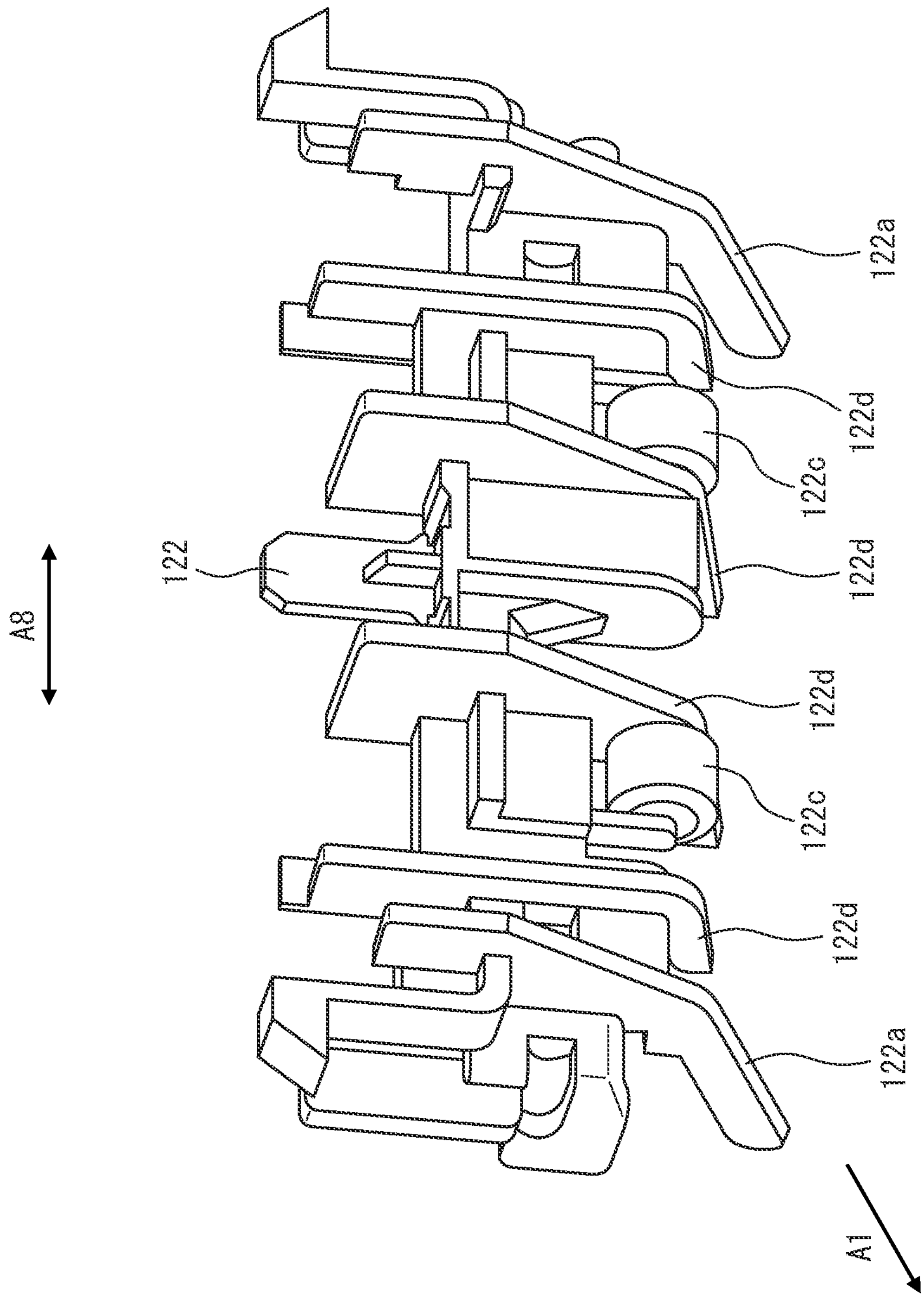


FIG. 12

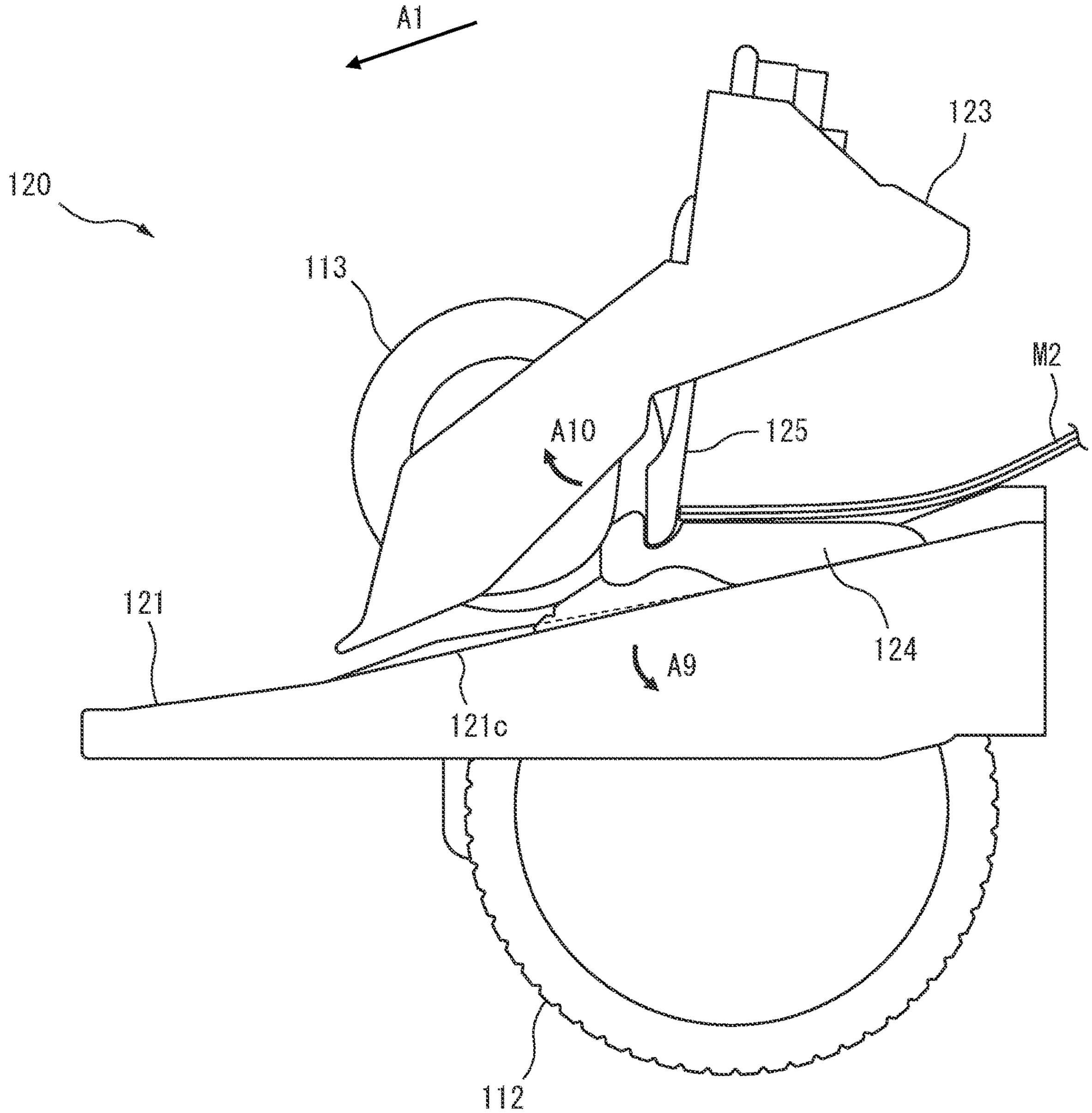


FIG. 13

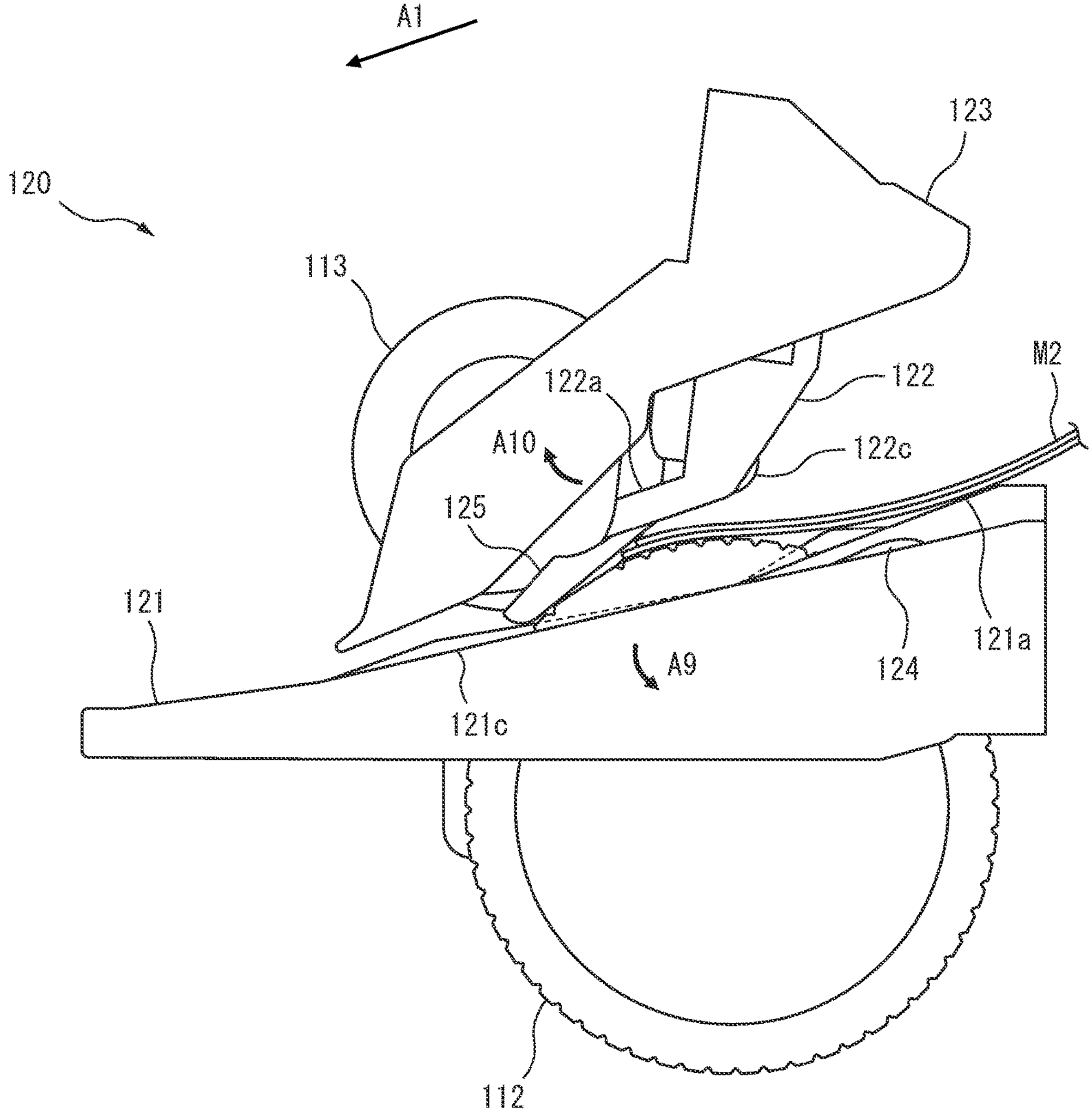


FIG. 14

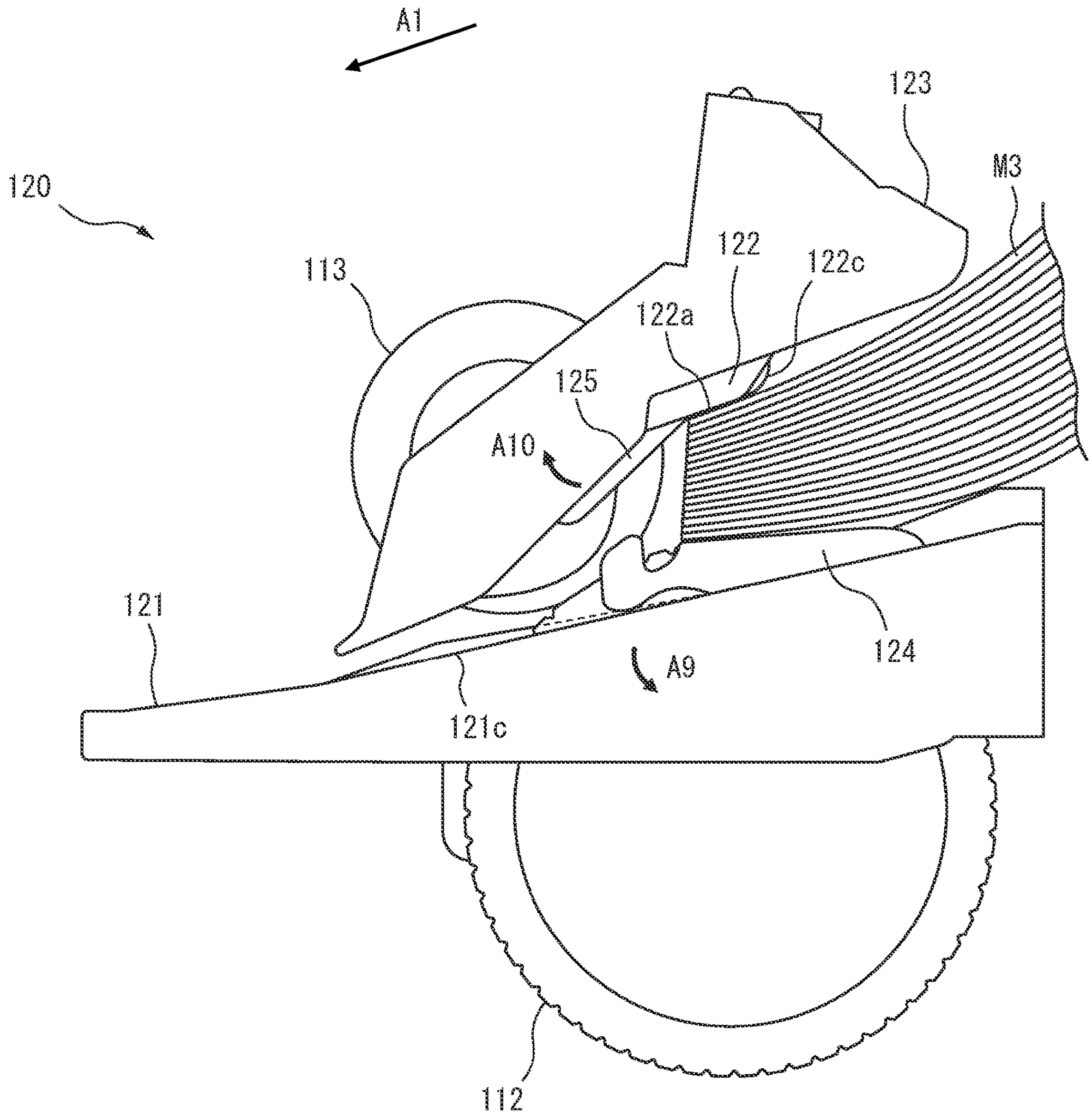


FIG. 15

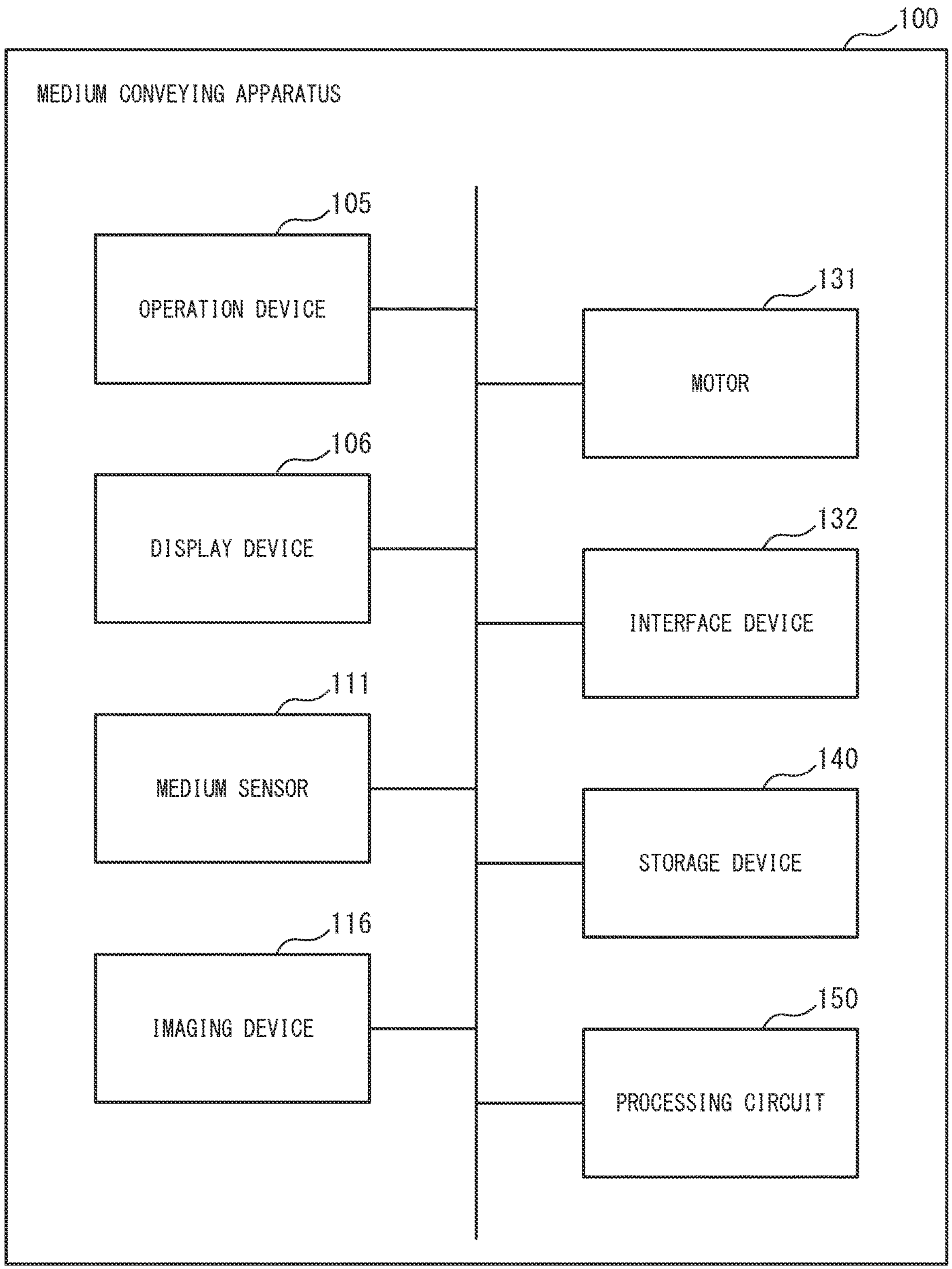

100
MEDIUM CONVEYING APPARATUS
105
OPERATION DEVICE
106
DISPLAY DEVICE
111
MEDIUM SENSOR
116
IMAGING DEVICE
131
MOTOR
132
INTERFACE DEVICE
140
STORAGE DEVICE
150
PROCESSING CIRCUIT

FIG. 16

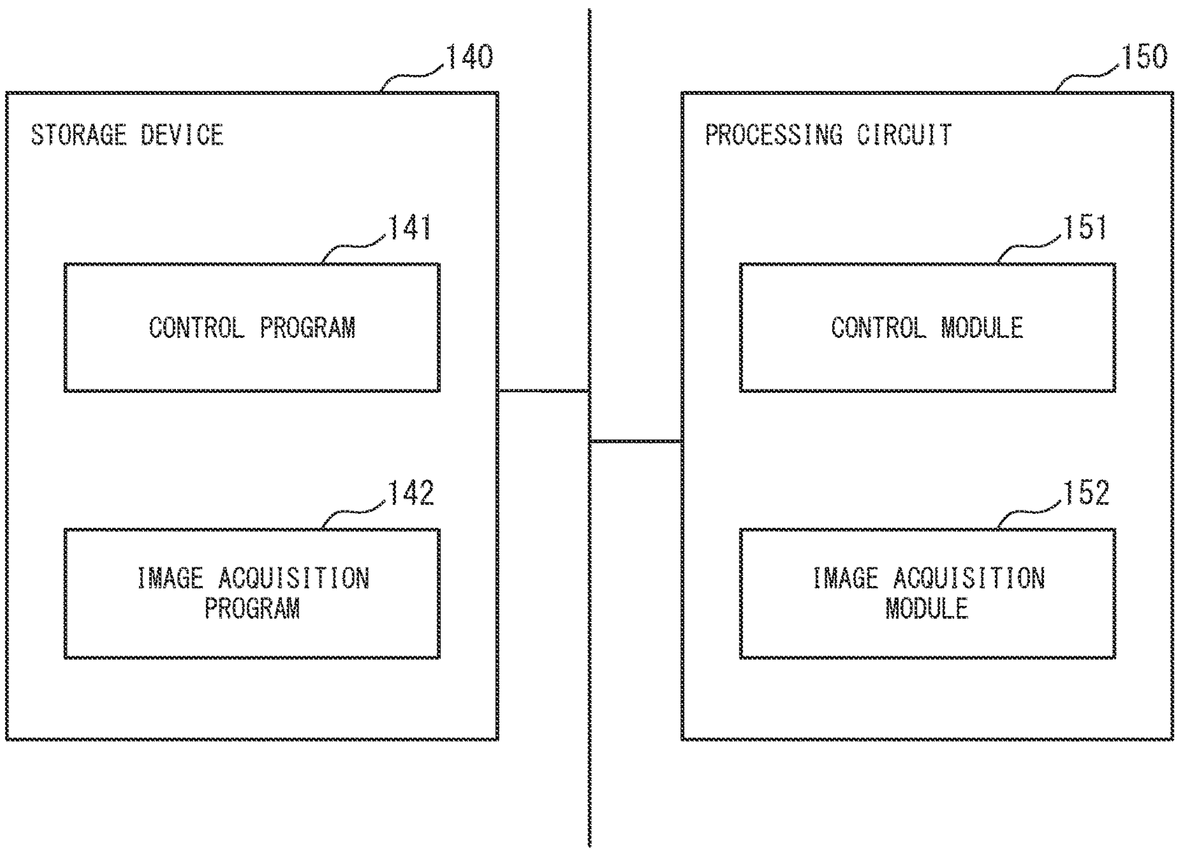

140
STORAGE DEVICE
141
CONTROL PROGRAM
142
IMAGE ACQUISITION PROGRAM
150
PROCESSING CIRCUIT
151
CONTROL MODULE
152
IMAGE ACQUISITION MODULE

FIG. 17

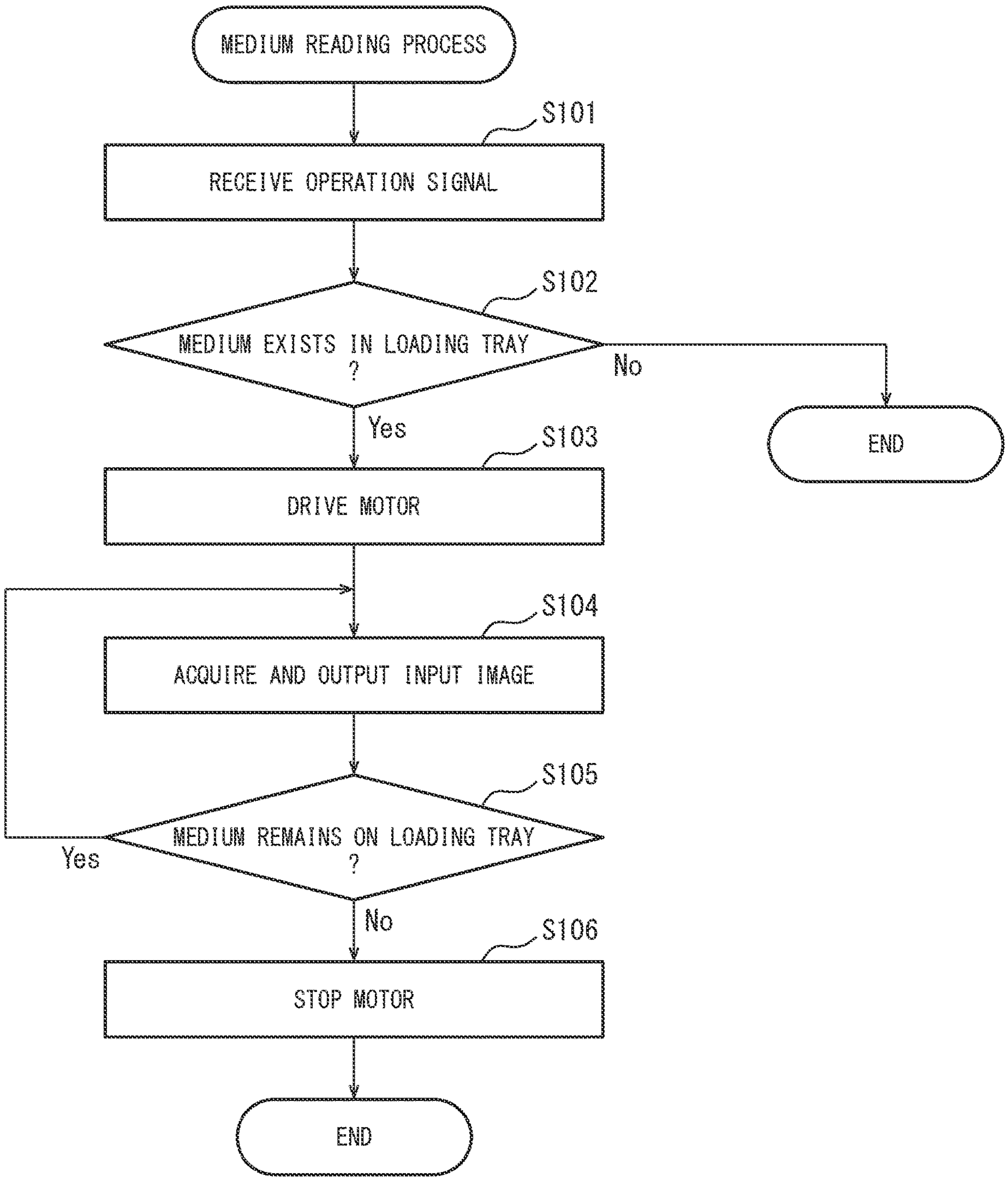

MEDIUM READING PROCESS
S101
RECEIVE OPERATION SIGNAL
S102
MEDIUM EXISTS IN LOADING TRAY ?
No
END
Yes
S103
DRIVE MOTOR
S104
ACQUIRE AND OUTPUT INPUT IMAGE
S105
MEDIUM REMAINS ON LOADING TRAY ?
Yes
No
S106
STOP MOTOR
END

FIG. 18

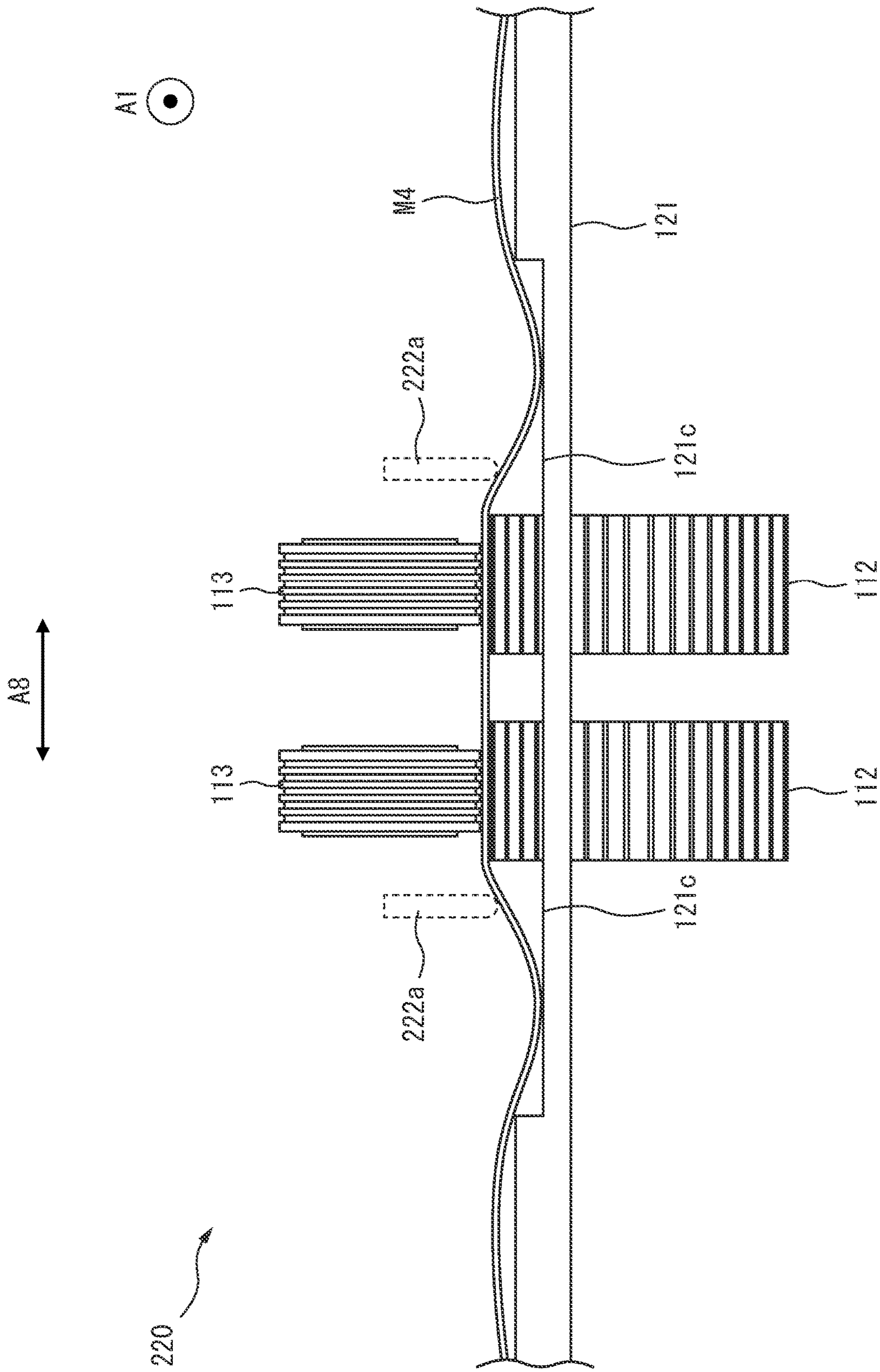


FIG. 19A

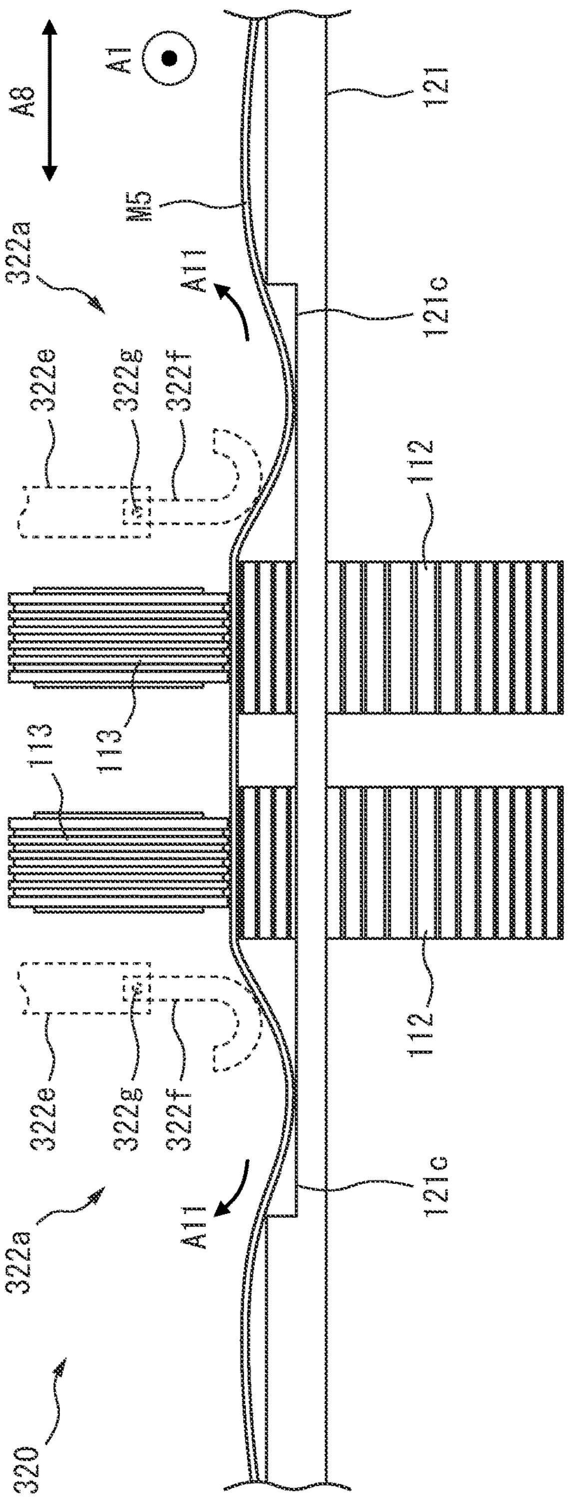
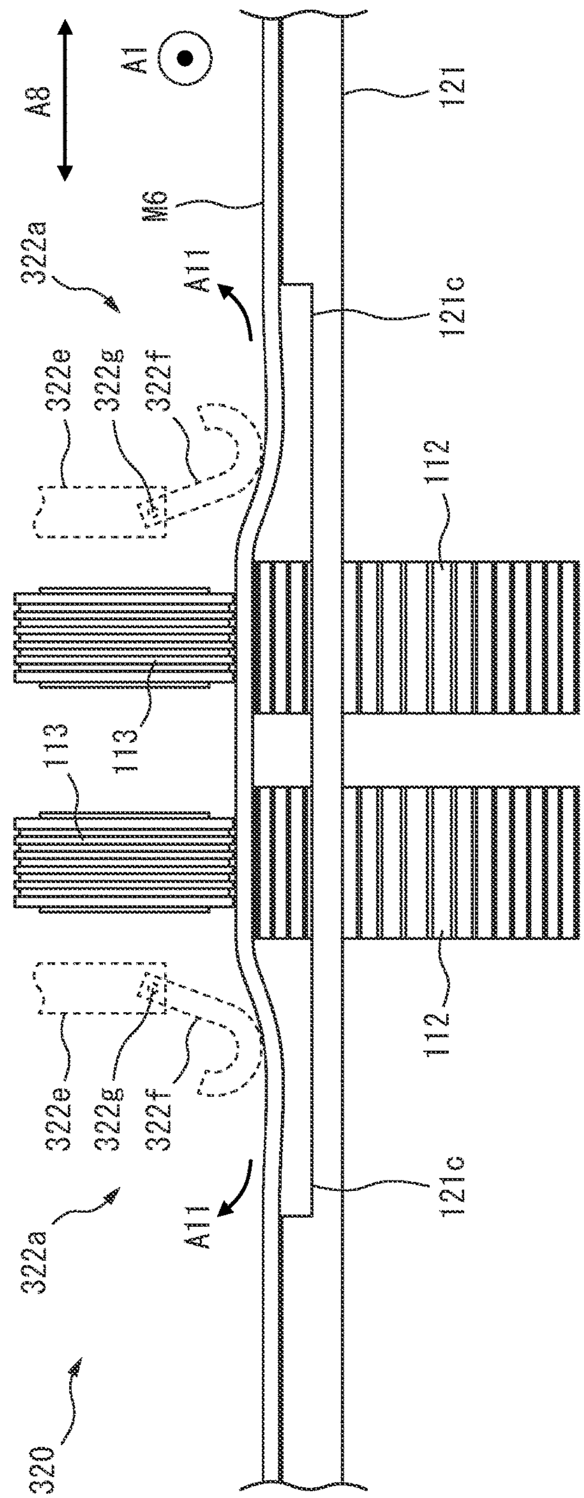


FIG. 19B



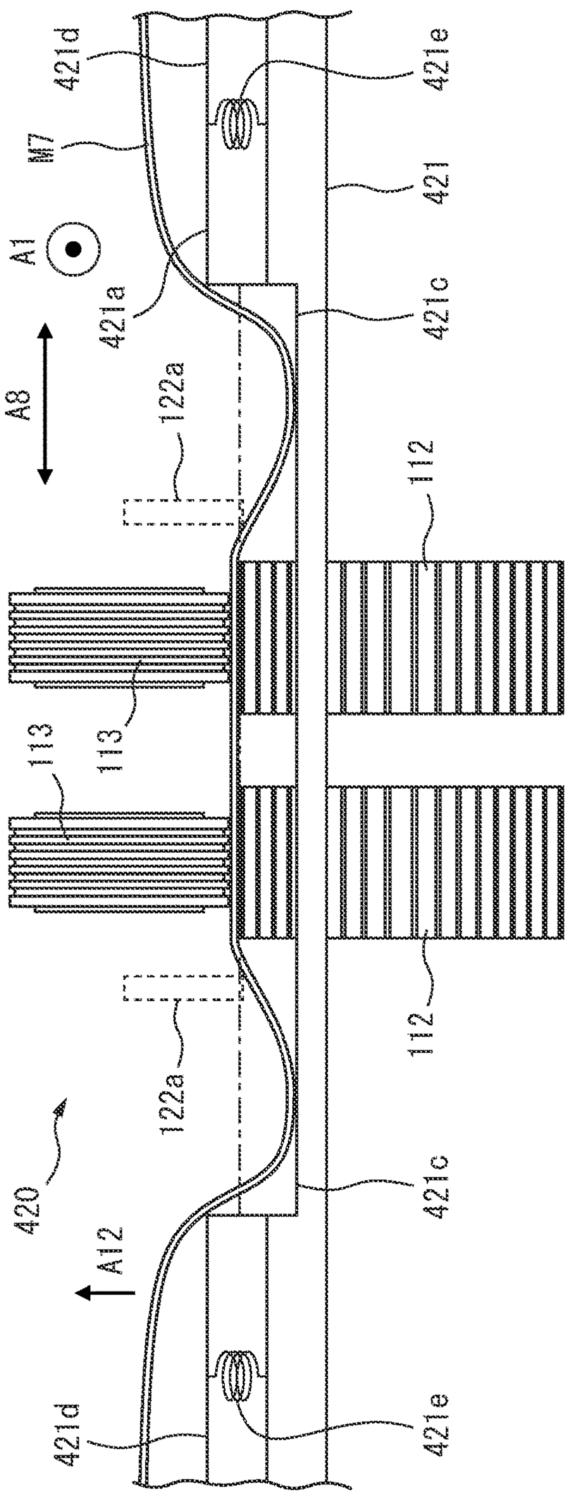


FIG. 20A

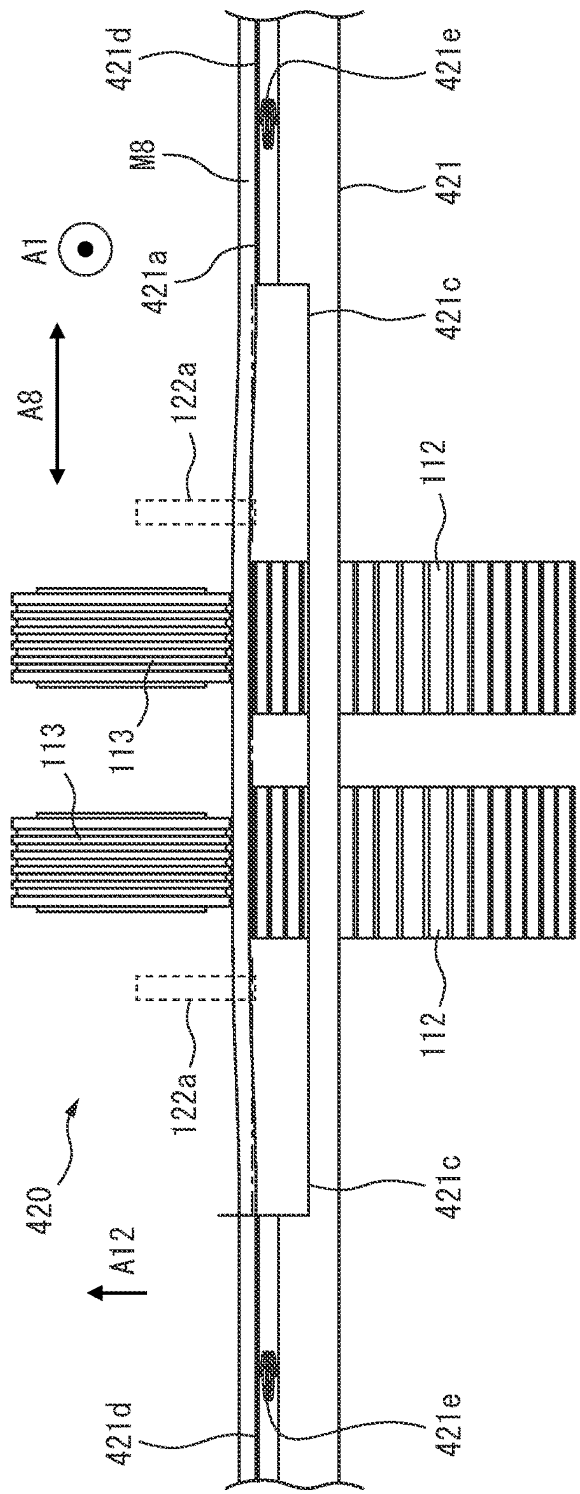


FIG. 20B

FIG. 21A

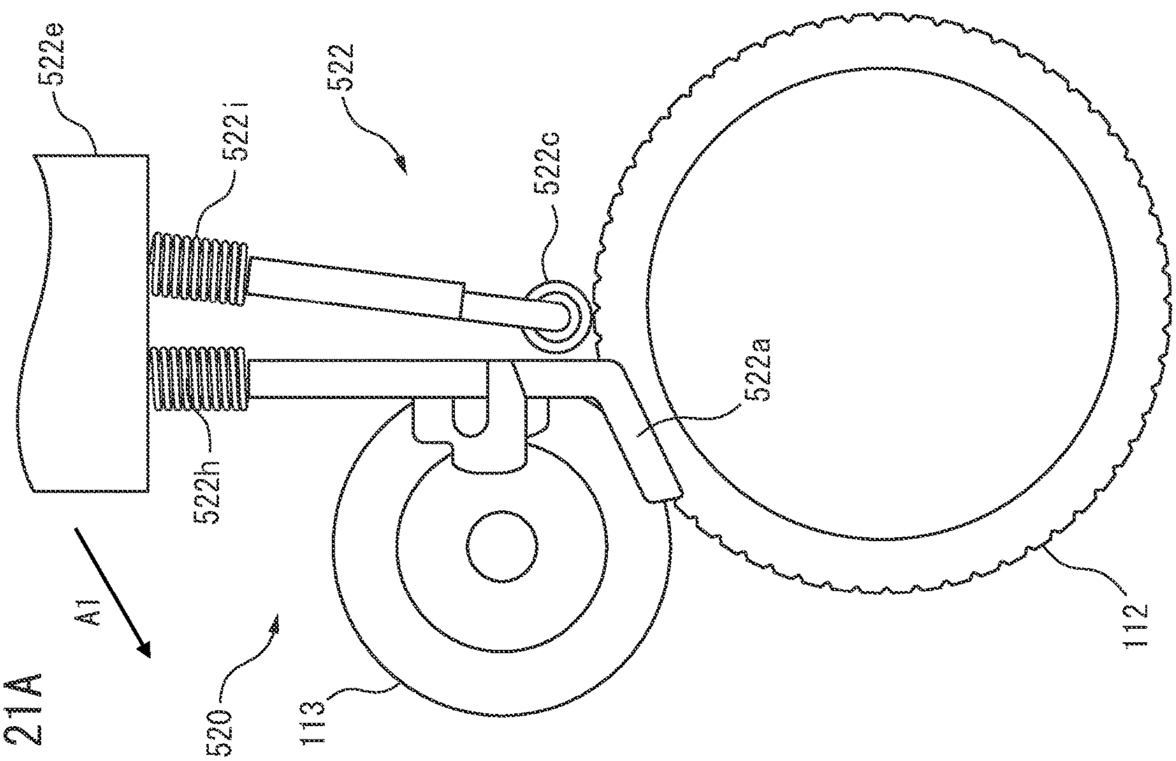


FIG. 21B

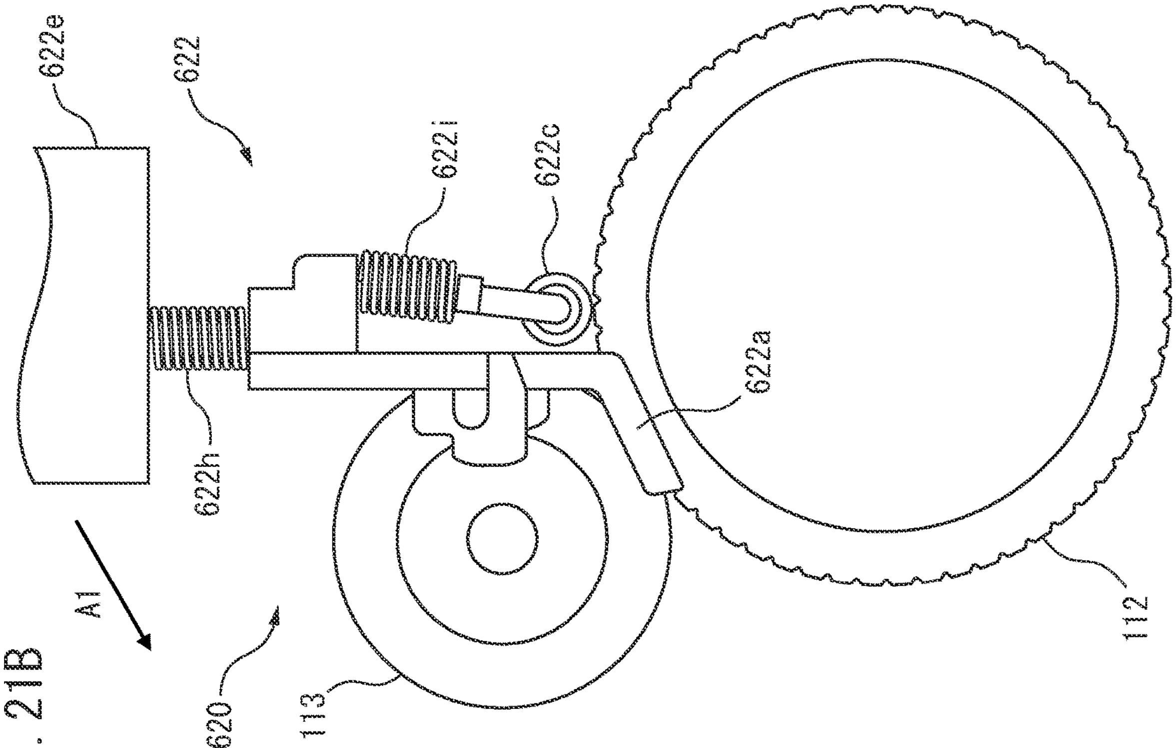


FIG. 22

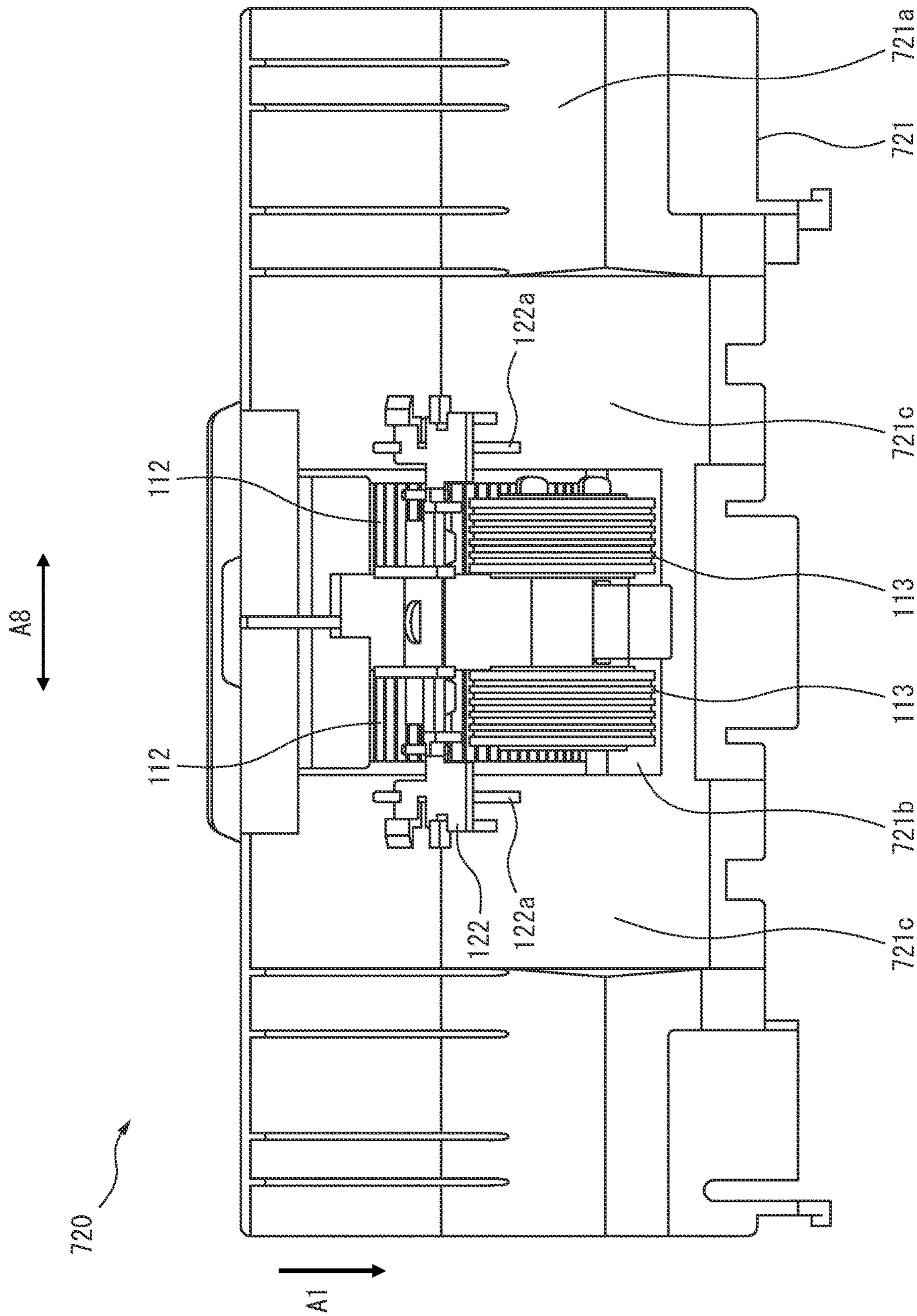


FIG. 23

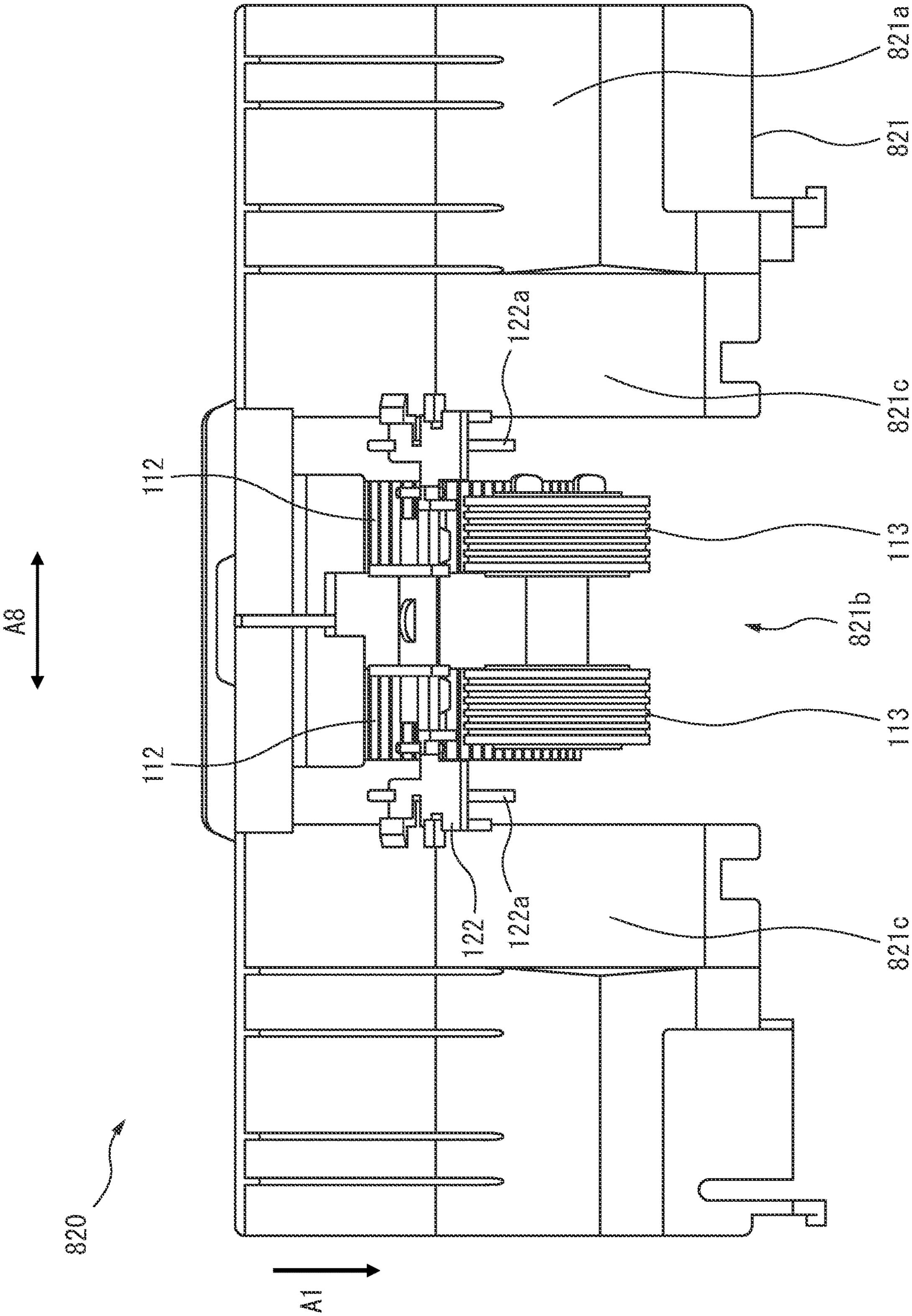


FIG. 24

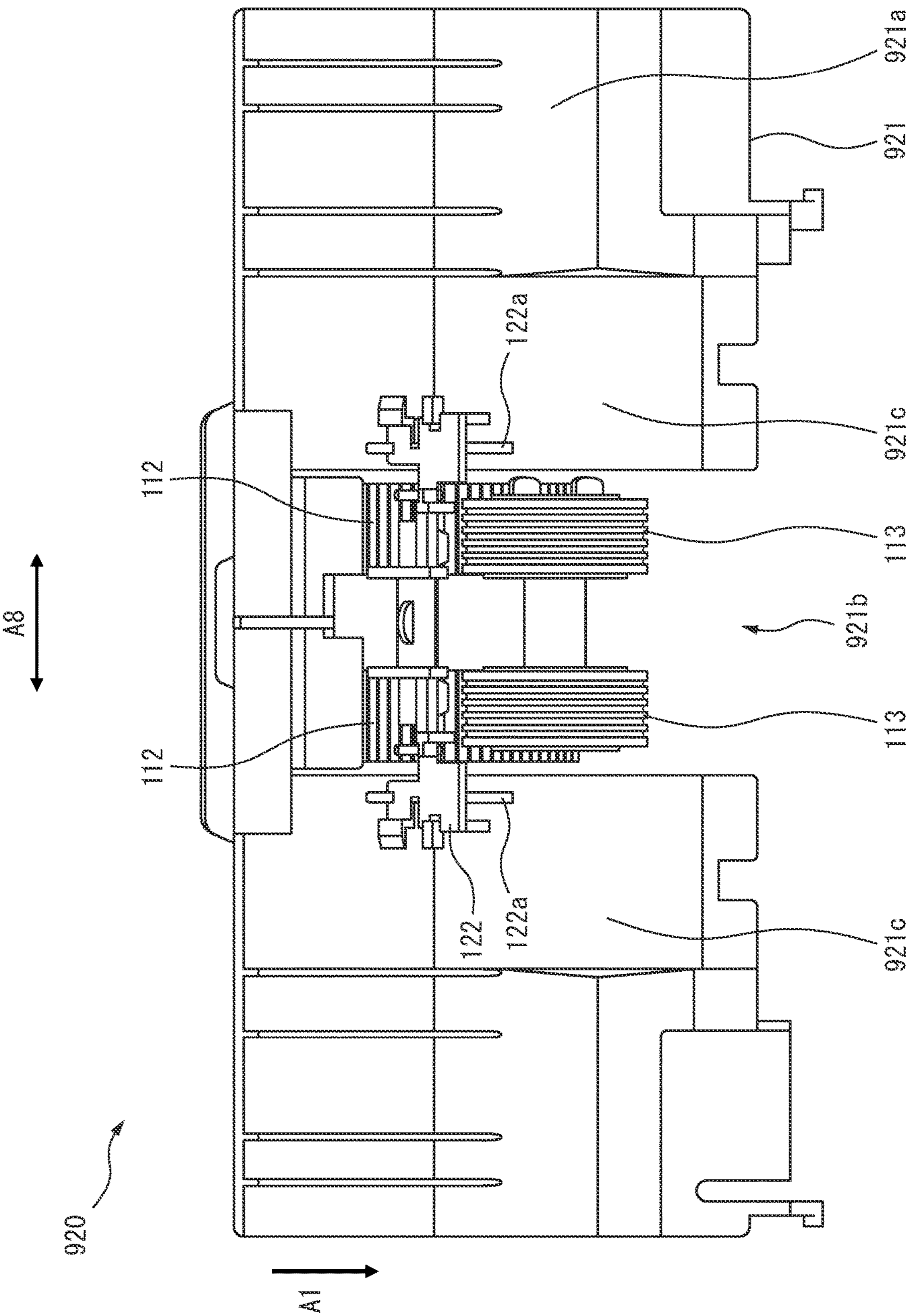
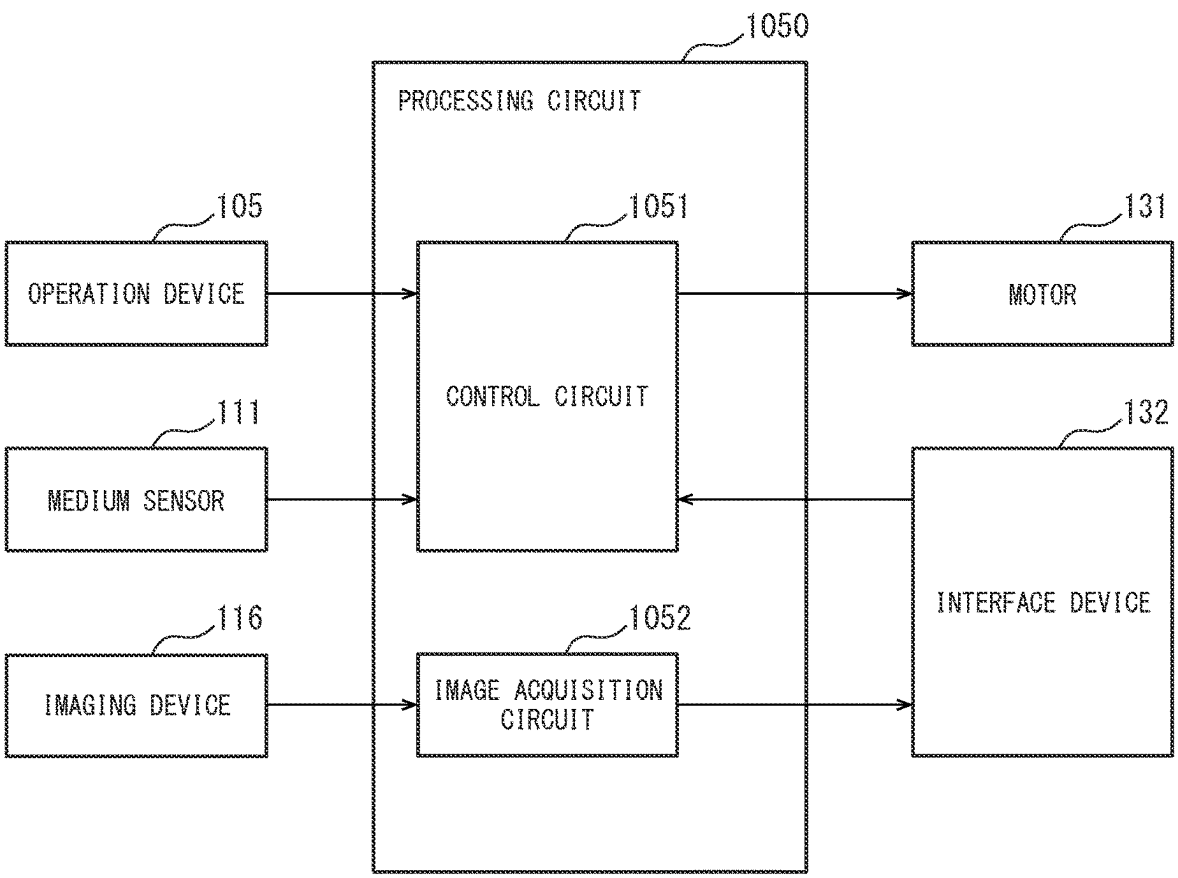


FIG. 25


1050
PROCESSING CIRCUIT
105
OPERATION DEVICE
1051
CONTROL CIRCUIT
131
MOTOR
111
MEDIUM SENSOR
132
INTERFACE DEVICE
116
IMAGING DEVICE
1052
IMAGE ACQUISITION CIRCUIT

MEDIUM CONVEYANCE DEVICE

CROSS-REFERENCE TO RELATED APPLICATION(S)

This application is a U.S. National Phase Patent Application and claims priority to and the benefit of International Application Number PCT/JP2021/022726, filed on Jun. 15, 2021 the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

The present disclosure relates to a medium conveying apparatus and particularly relates to a medium conveying apparatus including a feed roller and a separation roller, a control method, and a control program.

BACKGROUND ART

In recent years, it is preferred that a medium conveying apparatus to convey and image a plurality of media while separating the media, such as a scanner, convey not only common plain paper copier (PPC) paper but also various types of media, such as thin paper, an envelope, and bound carbon paper. Thin paper is likely to include a wrinkle or a tear and further is soft, and therefore when thin paper is conveyed as a medium, jamming of the medium due to occurrence of buckling of the medium is likely to occur. When a medium composed of a plurality of sheets of paper, such as an envelope or carbon paper, is conveyed, jamming of the medium may occur due to exertion of a separating force on the medium. A medium conveying apparatus generally has a separation mode of separating and feeding a medium and a non-separation mode of feeding a medium without separation as a feed mode. For example, by setting the feed mode of the medium conveying apparatus to the non-separation mode, occurrence of jamming of a medium is prevented; however, a user feels troublesomeness of having to change the feed mode for each medium type, which impairs user convenience. Further, when a plurality of types of media are collectively set in a loading tray and are sequentially conveyed, it is difficult to change the feed mode for each medium type.

A paper feeding apparatus including a paper feeding means for separating and feeding paper one sheet, assuming that the paper has a convexly curled state is disclosed (see PTL 1). The paper feeding apparatus includes a turn guide to reverse separated and fed paper in a convex direction and a guiding member provided between the turn guide and the paper feeding means, to guide paper reversed in the convex direction while adjusting the paper to a curled state in a direction opposite to the convex direction.

A paper feeding-conveyance apparatus including a paper feeding means, a conveyance roller pair to convey paper fed from the paper feeding means, and a lower paper feeding guide installed between the paper feeding means and the conveyance roller pair is disclosed (see PTL 2). The lower paper feeding guide includes a part where paper guide surfaces on both ends are higher than a guide surface in the central part.

CITATION LIST

Patent Literature

[PTL 1]
Japanese Unexamined Patent Publication (Kokai) No. H6-179540
[PTL 2]
Japanese Unexamined Patent Publication (Kokai) No. H2-305741

SUMMARY OF INVENTION

It is preferred that a medium conveying apparatus suitably feed a plurality of types of media.

An object of a medium conveying apparatus is to enable suitable feeding of a plurality of types of media.

According to some embodiments, a medium conveying apparatus includes a guide member including an opening or a notch and configured to form a conveyance surface of a medium, a feed roller located inside the opening or the notch and configured to feed the medium, a separation roller located to face the feed roller, and protruding parts located on left and right sides of the separation roller relative to a medium conveying direction. The guide member includes recessed parts positioned on left and right sides of the feed roller relative to the medium conveying direction. The protruding parts are located at positions facing the opening or the notch, or the recessed parts in such a way that the medium is conveyed between the protruding parts and the recessed parts. A tip of each of the protruding parts is positioned on the feed roller side of a nip part of the feed roller and the separation roller.

The medium conveying apparatus according to the present embodiment can suitably feed a plurality of types of media.

The object and advantages of the invention will be realized and attained by means of the elements and combinations, in particular, described in the claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory, and are not restrictive of the invention as claimed.

BRIEF DESCRIPTION OF DRAWINGS

FIG. **1** is a perspective view illustrating an example of a medium conveying apparatus according to an embodiment.

FIG. **2** is a diagram for illustrating a conveyance path inside the medium conveying apparatus.

FIG. **3** is a schematic diagram for illustrating an example of feeding mechanism.

FIG. **4** is a schematic diagram of the feeding mechanism viewed from above.

FIG. **5** is a schematic diagram of the feeding mechanism viewed from a downstream side.

FIG. **6** is a graph illustrating a relation between a length in a width direction A**8** and a buckling load for each medium.

FIG. **7** is an A-A' cross-sectional view of FIG. **4**.'

FIG. **8** is a schematic diagram illustrating a situation in which a medium M**1** is fed by the feeding mechanism.

FIG. **9** is a perspective view of an example of a feed arm etc., viewed from an upstream side.

FIG. **10** is a B-B' cross-sectional view of FIG. **4**.

FIG. **11** is a perspective view of the feed arm viewed from the upstream side.

FIG. **12** is a schematic diagram for illustrating an example of a setting guide etc.

FIG. **13** is a schematic diagram for illustrating the setting guide etc.

FIG. **14** is a schematic diagram for illustrating the setting guide etc.

FIG. **15** is a block diagram illustrating a schematic configuration of the medium conveying apparatus.

FIG. **16** is a diagram illustrating a schematic configuration of an example of a storage device and a processing circuit.

FIG. **17** is a flowchart illustrating an operation example of medium reading processing.

FIG. **18** is a schematic diagram for illustrating another example of protruding part.

FIG. **19**A is a schematic diagram for illustrating yet another example of protruding part.

FIG. **19**B is a schematic diagram for illustrating yet another example of protruding part.

FIG. **20**A is a schematic diagram for illustrating another example of guide member.

FIG. **20**B is a schematic diagram for illustrating another example of guide member.

FIG. **21**A is a schematic diagram for illustrating another example of feed arm.

FIG. **21**B is a schematic diagram for illustrating yet another example of feed arm.

FIG. **22** is a schematic diagram for illustrating yet another example of guide member.

FIG. **23** is a schematic diagram for illustrating yet another example of guide member.

FIG. **24** is a schematic diagram for illustrating yet another example of guide member.

FIG. **25** is a diagram illustrating a schematic configuration of another example of processing circuit.

DESCRIPTION OF EMBODIMENTS

Hereinafter, a medium conveying apparatus, a control method and a control program according to an embodiment, will be described with reference to the drawings. However, it should be noted that the technical scope of the invention is not limited to these embodiments, and extends to the inventions described in the claims and their equivalents.

FIG. **1** is a perspective view illustrating an example of a medium conveying apparatus configured as an image scanner. The medium conveying apparatus **100** conveys and images a medium being a document. For example, a medium is paper, thin paper, thick paper, a card, carbon paper, or an envelope. Carbon paper is a medium acquired by binding a plurality of sheets of thin paper, such as non-carbon paper. The medium conveying apparatus **100** may be a facsimile, a copying machine, a multifunctional peripheral (MFP), etc. A conveyed medium may be an object being printed on, etc., instead of a document, and the medium conveying apparatus **100** may be a printer, etc.

The medium conveying apparatus **100** includes a lower housing **101**, an upper housing **102**, a loading tray **103**, an ejection tray **104**, an operation device **105**, a display device **106**, etc.

The upper housing **102** is located at a position covering the top surface of the medium conveying apparatus **100** and is engaged with the lower housing **101** by a hinge in such a way as to be openable when, for example, a medium is stuck or cleaning of the inside of the medium conveying apparatus **100** is performed.

The loading tray **103** is engaged with the lower housing **101** and places a medium to be fed and conveyed. The ejection tray **104** is engaged with the upper housing **102** and places an ejected medium. The ejection tray **104** may be engaged with the lower housing **101**.

The operation device **105** includes an input device such as a button, and an interface circuit acquiring a signal from the input device, accepts an input operation by a user, and outputs an operation signal based on the input operation by the user. The display device **106** includes a display including a liquid crystal, an organic electro-luminescence (EL), etc., and an interface circuit outputting image data to the display, and displays the image data on the display.

FIG. **2** is a diagram for illustrating a conveyance path inside the medium conveying apparatus.

The conveyance path inside the medium conveying apparatus **100** includes a medium sensor **111**, a feed roller **112**, a separation roller **113**, a first conveyance roller **114**, a second conveyance roller **115**, an imaging device **116**, a first ejection roller **117**, a second ejection roller **118**, etc.

Each of the numbers of the feed roller **112**, the separation roller **113**, the first conveyance roller **114**, the second conveyance roller **115**, the first ejection roller **117**, and/or the second ejection roller **118** is not limited to one and may be more than one. In that case, a plurality of feed rollers **112**, separation rollers **113**, first conveyance rollers **114**, second conveyance rollers **115**, first ejection rollers **117**, and/or second ejection rollers **118** are respectively spaced in the width direction A**2** perpendicular to a medium conveying direction.

The top surface of the lower housing **101** forms a lower guide **101***a* of the conveyance path of a medium, and the bottom surface of the upper housing **102** forms an upper guide **102***a* of the conveyance path of a medium. An arrow A**1** in FIG. **2** indicates a medium conveying direction. Hereinafter, an upper stream refers to an upper stream in the medium conveying direction A**1**, and a lower stream refers to a lower stream in the medium conveying direction A**1**.

The medium sensor **111** is located on the upstream side of the feed roller **112** and the separation roller **113**. The medium sensor **111** includes a contact detection sensor and detects whether a medium is placed in the loading tray **103**. The medium sensor **111** generates and outputs a first medium signal the signal value of which varies between a state in which a medium is placed in the loading tray **103** and a state in which a medium is not placed. The medium sensor **111** is not limited to a contact detection sensor and any other sensor that can detect existence of a medium, such as a light detection sensor, may be used as the medium sensor **111**.

The feed roller **112** is provided in the lower housing **101**, sequentially separates media placed in the loading tray **103** from the lower side, and feeds the media. The separation roller **113** is a so-called brake roller or retard roller, is provided in the upper housing **102** to face the feed roller **114**, and rotates in a direction opposite to the medium feeding direction. The feed roller **112** may be provided in the upper housing **102**, the separation roller **113** may be provided in the lower housing **102**, and the feed roller **112** may sequentially separates media placed in the loading tray **103** from the upper side.

The first conveyance roller **114** and the second conveyance roller **115** are located on the downstream side of the feed roller **112** to face each other and convey a medium fed by the feed roller **112** and the separation roller **113** to the imaging device **116**. The first conveyance roller **114** is provided in the lower housing **101**, and the second conveyance roller **115** is provided in the upper housing **102** and above the first conveyance roller **114**.

The imaging device **116** is located on the downstream side of the first conveyance roller **114** and images a medium conveyed by the first conveyance roller **114**. The imaging device **116** includes a first imaging device **116***a* and a second imaging device **116***b* that are located to face each other with the medium conveyance path in between. The first imaging device **116***a* includes a line sensor based on a unity-magnification optical system type contact image sensor (CIS) including complementary metal oxide semiconductor-(CMOS-) based imaging elements linearly arranged in a main scanning direction. The first imaging device **118***a* further includes lenses each forming an image on an imaging element, and an A/D converter amplifying and analog-digital (A/D) converting an electric signal output from the imaging element. The first imaging device **116***a* generates an input image by imaging the front side of a conveyed medium in accordance with control from a processing circuit to be described later and outputs the generated image.

Similarly, the second imaging device **116***b* includes a line sensor based on a unity-magnification optical system type CIS including CMOS-based imaging elements linearly arranged in the main scanning direction. The second imaging device **116***b* further includes lenses each forming an image on an imaging element, and an A/D converter amplifying and analog-digital (A/D) converting an electric signal output from the imaging element. The second imaging device **116***b* generates an input image by imaging the back side of a conveyed medium in accordance with control from the processing circuit to be described later and outputs the generated image.

Only one of the first imaging device **116***a* and the second imaging device **116***b* may be located and only one side of a medium may be read in the medium conveying apparatus **100**. Further, a line sensor based on a unity-magnification optical system type CIS including charge coupled device-(CCD-) based imaging elements may be used in place of the line sensor based on a unity-magnification optical system type CIS including CMOS-based imaging elements. Further, a reduction optical system type line sensor including CMOS-based or CCD-based imaging elements may be used.

The first ejection roller **117** and the second ejection roller **118** are located on the downstream side of the imaging device **116** to face each other and eject a medium conveyed by the first conveyance roller **114** and the second conveyance roller **115** and imaged by the imaging device **116** into the ejection tray **104**. The first ejection roller **117** is provided in the lower housing **101**, and the second ejection roller **118** is provided in the upper housing **102** and above the first ejection roller **119**.

A medium placed in the loading tray **103** is conveyed between the lower guide **101***a* and the upper guide **102***a* toward the medium conveying direction A1 by the feed roller **112** rotating in a direction of an arrow A2 in FIG. **2**, i.e., the medium feeding direction. As a feed mode, the medium conveying apparatus **100** has a separation mode of separating and feeding a medium and a non-separation mode of feeding a medium without separation. The feed mode is set by a user by using the operation device **105** or an information processing apparatus connected to the medium conveying apparatus **100** by communication. When the feed mode is set to the separation mode, the separation roller **113** rotates in a direction of an arrow A3, i.e., a direction opposite to the medium feeding direction at the time of medium feeding. When a plurality of media are placed in the loading tray **103**, only a medium in contact with the feed roller **112** out of the medium placed in the loading tray **103** is separated by working of the feed roller **112** and the separation roller **115**. Consequently, conveyance of a medium other than the separated medium is restricted (prevention of multi feed). On the other hand, when the feed mode is set to the non-separation mode, the separation roller **113** rotates in a direction opposite to the arrow A**3**, i.e., the medium feeding direction.

A medium is fed between the first conveyance roller **114** and the second conveyance roller **115** while being guided by the lower guide **101***a* and the upper guide **102***a*. The medium is fed between the first imaging device **116***a* and the second imaging device **116***b* by the first conveyance roller **114** and the second conveyance roller **115** rotating in directions of an arrow A4 and an arrow A5, respectively. The medium read by the imaging device **116** is ejected into the ejection tray **104** by the first ejection roller **117** and the second ejection roller **118** rotating in directions of an arrow A**6** and an arrow A**7**, respectively.

FIG. **3** is a schematic diagram for illustrating an example of a feeding mechanism in the medium conveying apparatus.

As illustrated in FIG. **3**, the medium conveying apparatus **100** includes a guide member **121** and a feed arm **122** in addition to the feed roller **112** and the separation roller **113** as the feeding mechanism **120**. In the example illustrated in FIG. **3**, the medium conveying apparatus **100** includes two each of the feed rollers **112** and the separation rollers **113**.

The guide member **121** is a plate-shaped member, is provided on the top surface of the lower housing **101** to form a conveyance surface **121***a* of a medium and forms part of the lower guide **101***a*. The guide member **121** has an opening **121***b* in the central part in the width direction A**2** perpendicular to the medium conveying direction, and the feed roller **112** is located in the opening **121***b*.

The conveyance surface **121***a* of the guide member **121** includes recessed parts **121***c* positioned on the left and right sides of the feed roller **112** relative to the medium conveying direction A**1**. In the example illustrated in FIG. **3**, the guide member **121** includes two recessed parts **121***c*. The recessed part **121***c* is formed in such a way that the downstream side of the recessed part **121***c* is inclined downward relative to a region **121***d* in the conveyance surface **121***a* on the upstream side of the recessed part **121***c*

The feed arm **122** is provided in the upper housing **102** and is located on the upstream side of the separation roller **113** and in the neighborhood of the separation roller **113**. The feed arm **122** includes protruding parts **122***a* extending from the upper housing **102** side to the lower housing **101** side, i.e., from the separation roller **113** side to the feed roller **112** side. The protruding parts **122***a* are located on the left and right sides of the separation roller **113** relative to the medium conveying direction A**1**. In particular, the protruding parts **122***a* are located at positions facing the opening **121***b* of the guide member **121**. In the example illustrated in FIG. **3**, the feed arm **122** includes two protruding parts **122***a*.

FIG. **4** is a schematic diagram of the feeding mechanism viewed from above.

As illustrated in FIG. **4**, the recessed part **121***c* is provided in a region R1 being in the central part and being outside the opening **121***b* in the width direction A**8**. Further, the recessed part **121***c* is provided in a region R**2** including a nip part of the feed roller **112** and the separation roller **113** in the medium conveying direction A**1**. While the upstream edge P3 of the recessed part **121***c* is positioned on the downstream side of the upstream edge of the feed roller **112** in the example illustrated in FIG. **4**, the recessed part **121***c* may be provided in a region including the entire feed roller **112** and the entire separation roller **113** in the medium conveying direction A**1**.

FIG. **5** is a schematic diagram of the feeding mechanism viewed from the downstream side. In FIG. **5**, display of parts other than the protruding part **122***a* of the feed arm **122** is omitted for enhanced visual recognizability.

As illustrated in FIG. **5**, a tip (lower end) of the protruding part **122***a* is positioned on the feed roller **112** side of (below) the nip part of the feed roller **112** and the separation roller **113**. A medium M**1** fed by the feed roller **112** and the separation roller **113** is conveyed between the protruding part **122***a* of the feed arm **122** and the recessed part **121***c* of the guide member **121**.

The protruding part **122***a* pushes down a region of the fed medium M**1** facing the recessed part **121***c* in the width direction A**8** below than an inner region in contact with the feed roller **112** and the separation roller **113**, and a region outside the recessed part **121***c*. Consequently, the fed medium M**1** is wavily bent in the width direction A**8**, and therefore the medium conveying apparatus **100** can stiffen the medium and can improve stiffness of the medium moving forward in the medium conveying direction A**1**. Accordingly, even when soft thin paper is conveyed as a medium, the medium conveying apparatus **100** can suppress occurrence of buckling of the medium and suppress occurrence of jamming of the medium. Further, even when a medium composed of a plurality of sheets of paper, such as an envelope or carbon paper, is conveyed, the medium is provided with stiffness that can resist a separating force by the separation roller **113**; and therefore the medium conveying apparatus **100** can suppress occurrence of jamming of the medium.

The protruding part **122***a* pushes down the region in the medium M**1** facing the recessed part **121***c* in the width direction A**8** below than the inner region in contact with the feed roller **112** and the separation roller **113**. Consequently, the position in a height direction perpendicular to the conveyance surface **121***a* varies between the inner region in contact with the feed roller **112** and the separation roller **113**, and a region outside the inner region in the medium M**1**. Therefore, when a plurality of media are collectively set in the loading tray **103** and are fed, the frictional force between the medium placed between the feed roller **112** and the separation roller **113** is less likely to propagate outward in the width direction A**8**. Accordingly, the medium conveying apparatus **100** can suppress occurrence of jamming of a special medium, such as thin paper, an envelope, or carbon paper, while improving separation performance for normal paper, such as PPC paper.

A region outside the recessed part **121***c* in the width direction A**8** in the conveyance surface **121***a* of the guide member **121** is positioned on the feed roller **112** side of (below) the nip part of the feed roller **112** and the separation roller **113**. Assuming that the conveyance surface **121***a* is located above the nip part, a medium is less likely to come in contact with the feed roller **112**, and the medium conveyance force is reduced. By the conveyance surface **121***a* located below the nip part, the medium conveying apparatus **100** can satisfactorily stiffen a medium while suppressing reduction in medium conveyance force.

A gap is provided between the tip of the protruding part **122***a* and the recessed part **121***c* in the height direction perpendicular to the conveyance surface **121***a*. Consequently, the medium is satisfactorily fed between the protruding part **122***a* and the recessed part **121***c* without being hampered by the protruding part **122***a* and the recessed part **121***c*.

The distance W [mm] from the outer edge of the feed roller **112** to the outer edge of the recessed part **121***c* is preferably set to satisfy Equation (1) below.

$$W=(Wc-Wr)/2>10\text{ [mm]} \tag{1}$$

Wc denotes the length [mm] in the width direction A**8** between both ends of the recessed parts **121***c*, and Wr denotes the length [mm] in the width direction A**8** between both ends of the feed rollers **112**. The length Wr in the width direction A**8** between both ends of the feed rollers **112** is set to a length less than or equal to the minimum medium size width supported by the medium conveying apparatus **100** (such as the length of the A**8** size in the widthwise direction).

The length We in the width direction A**8** between both ends of the recessed parts **121***c* is preferably set to satisfy Equation (2) below.

$$(Wm-Wc)/2>10\text{ [mm]} \tag{2}$$

Wm denotes the minimum medium size width of a mainly fed medium and, for example, is set to 148 [mm] being the length of the A**5** size in the widthwise direction (the length of the A**6** size in the lengthwise direction). Consequently, both ends of the fed medium are placed outside the recessed parts **121***c* in the width direction A**8**, and the outer region of the medium is positioned above a region facing the recessed parts **121***c*; and therefore the medium conveying apparatus **100** can satisfactorily bend the medium.

The buckling load T [gf] required for a fed medium for buckling is calculated by Equation (3) below

$$T=\pi^2\times E\times M/L^2 \tag{3}$$

E denotes the Young's modulus [GPa] of the medium. L denotes the length [mm] in the width direction A**8** of a region in the medium to which the load is applied. M denotes a sectional secondary moment. The sectional secondary moment M is a characteristic of a section indicating resistance to deformation of a member against a bending force and is calculated by Equation (4) below.

$$M=H^3\times B/12 \tag{4}$$

H denotes the thickness [mm] of the medium, and B denotes the length [mm] in the medium conveying direction A**1** of the region in the medium to which the load is applied.

FIG. **6** is a graph illustrating a relation between the length in the width direction A**8** and the buckling load for each medium.

In FIG. **6**, a graph G**1** illustrates the buckling load of non-carbon paper with a Young's modulus of 3 [GPa], and a graph G**2** illustrates the buckling load of thin paper with a Young's modulus of 2.2 [GPa]. In FIG. **6**, the horizontal axis indicates the length [mm] in the width direction A**8** of each medium, and the vertical axis indicates the buckling load [gf] of each medium. The length in the medium conveying direction A**1** of a region in each medium indicated in FIG. **6** to which the load is applied is assumed to be 40 [mm]. As indicated in Equation (3), the buckling load required for buckling a medium is inversely proportional to the square of the length in the width direction **A8** of the region in the medium to which the load is applied. As illustrated in FIG. **6**, when the length in the width direction A**8** of a medium is less than or equal to 10 [mm], the buckling load required for buckling the medium sharply increases.

Accordingly, as indicated in Equation (1), by setting the distance W from the outer edge of the feed roller **112** to the outer edge of the recessed part **121***c* greater than 10 [mm], the medium conveying apparatus **100** can satisfactorily buckle a medium with a sufficiently light buckling load. Consequently, the medium conveying apparatus **100** can buckle a region in a medium facing the recessed part **121***c* by the protruding part **122***a* and the self-weight of the medium without using a special pressing member and can satisfactorily stiffen the medium.

The length Wa in the width direction A**8** between both ends of the two protruding parts **122***a* is set to a length greater than the length Wr in the width direction A**8** between both ends of the feed rollers **112** and less than or equal to the minimum medium size width supported by the medium conveying apparatus **100**. In other words, the two protruding parts **122***a* are located outside the two feed rollers **112** and in the neighborhood of the two feed rollers **112**. Consequently, the protruding part **122***a* can reliably stiffen small-sized paper, such as the A5 size, and the medium conveying apparatus **100** can suppress occurrence of jamming of a small-sized medium.

FIG. **7** is an A-A' cross-sectional view of FIG. **4**. In other words, FIG. **7** is a diagram of the feeding mechanism viewed from the side (a diagram viewed from the width direction A**8**)

As illustrated in FIG. **7**, the separation roller **113** is supported by a separation roller cover **123**. The separation roller cover **123** is mounted on the upper housing **102** through an elastic member, such as a spring or a rubber, and is urged downward by the elastic member. Consequently, the separation roller cover **123** provides an urging force to the separation roller **113** in such a way that the separation roller **113** presses the feed roller **112**.

The feed arm **122** is stored in the separation roller cover **123** in such a way as to be movable in the vertical direction relative to the separation roller cover **123**. The feed arm **122** is mounted in the separation roller cover **123** through an elastic member, such as a spring or a rubber, and is urged downward relative to the separation roller cover **123** by the elastic member.

A contact surface **122***b* of the protruding part **122***a* of the feed arm **122**, the surface coming in contact with a medium fed by the feed roller **112**, is formed in such a way as to be in parallel with the region **121***d* on the upstream side of the recessed part **121***c* of the conveyance surface **121***a* of the guide member **121** in the medium conveying direction A**1**. The contact surface **122***b* being in parallel with the region **121***d* is not limited to being completely in parallel and means that an angle formed by the contact surface **122***b* and the region **121***d* is less than or equal to a predetermined angle (such as 5°). By the contact surface **122***b* being formed in parallel with the region **121***d*, the protruding part **122***a* can smoothly guide a medium guided by the conveyance surface **121***a* of the guide member **121** toward the downstream side.

Assuming that the contact surface **122***b* is tilted in such a way that the downstream side of the contact surface **122***b* sinks relative to the region **121***d* (toward the feed roller **112** side), the magnitude of the load received by a medium from the protruding part **122***a* gradually increases as the medium is conveyed and the medium conveyance force decreases. In that case, when the medium is conveyed in a tilted manner, the magnitude of the load received by the medium from the left protruding part **122***a* differs from that of the load received from the right protruding part **122***a*, and therefore the tilt of the medium increases. On the other hand, when the contact surface **122***b* is tilted in such a way that the downstream side of the contact surface **122***b* floats relative to the region **121***d* (toward the separation roller **113** side), the force provided by the protruding part **122***a* for stiffening the medium decreases. By the contact surface **122***b* provided in parallel with the region **121***d*, the medium conveying apparatus **100** can satisfactorily stiffen the medium while suppressing reduction in the medium conveyance force and occurrence of a skew of the medium.

The protruding part **122***a* is located in such a way that, viewed from the width direction **A8** perpendicular to the medium conveying direction A**1**, at least part of the protruding part **122***a* overlaps the nip part of the feed roller **112** and the separation roller **113**, and a region of the feed roller **112** on the upstream side of the nip part in the medium conveying direction A**1**. In other words, the upstream edge P**1** of the protruding part **122***a* is located on the downstream side of the upstream edge of the feed roller **112** and on the upstream side of the upstream edge of the nip part of the feed roller **112** and the separation roller **113** in the medium conveying direction A**1**. Consequently, stiffening is performed on a fed medium by the protruding part **122***a* before the nip part of the feed roller **112** and the separation roller **113**. Accordingly, the medium conveying apparatus **100** can improve stiffness of a medium and can suppress occurrence of jamming of the medium when the medium enters the nip part. Further, even when the front edge of a fed medium is curled upward, the curled part is pressed down by the protruding part **122***a*, and therefore the medium is satisfactorily guided to the nip part of the feed roller **112** and the separation roller **113**.

The downstream edge P**2** of the protruding part **122***a* is located on the downstream side of a straight line LI connecting the center O**1** of the feed roller **112** and the center O**2** of the separation roller **113** and on the upstream side of the downstream edge of the feed roller **112** in the medium conveying direction A**1**. Further, the edge P**2** is located in the neighborhood of the downstream edge of the nip part of the feed roller **112** and the separation roller **113** in the medium conveying direction A**1**. In particular, the edge P**2** is located on the upstream side of the downstream edge of the nip part in the medium conveying direction A**1**. The edge P**2** may be located on the downstream side of the downstream edge of the nip part in the medium conveying direction A**1**. In general, jamming of a medium is likely to occur in a period from immediately before entry of the medium into the nip part until the medium passes the nip part. By the protruding part **122***a* extending to the neighborhood of the downstream edge of the nip part, stiffening is performed on the medium by the protruding part **122***a* in the neighborhood of the nip part. Accordingly, the medium conveying apparatus **100** can improve stiffness of the medium and can suppress occurrence of jamming of the medium while the medium is passing the nip part.

The protruding part **122***a* is located in such a way that at least part of the protruding part **122***a* is positioned below the upper edge of the feed roller **112** in the height direction perpendicular to the conveyance surface **121***a*. The bottom surface of the protruding part **122***a* is located in such a way as to be positioned below the upper edge of the feed roller

112 by a predetermined distance TI (for example, greater than or equal to 0.1 mm and less than or equal to 5 mm) or greater.

FIG. **8** is a schematic diagram illustrating a situation in which a medium M**1** is fed by the feeding mechanism. FIG. **8** is a schematic diagram of the feeding mechanism **120** viewed from the side.

As illustrated in FIG. **8**, by the outer part of the fed medium M**1** in the width direction A**8** pushed down by the protruding part **122***a*, a part of the fed medium M**1** facing the feed roller **112** extends along the surface of the feed roller **112**. Consequently, the medium M**1** is wavily bent in the medium conveying direction A**1**, and therefore the medium conveying apparatus **100** can stiffen the medium M**1** and can improve stiffness of the medium M**1** moving forward in the medium conveying direction A**1**. Accordingly, even when thin paper, an envelope, or carbon paper is conveyed as a medium, the medium conveying apparatus **100** can suppress occurrence of jamming of the medium.

FIG. **9** is a perspective view of the feed arm, the feed roller, and the separation roller viewed from the upstream side. FIG. **10** is a B-B' cross-sectional view of FIG. **4**. In other words, FIG. **10** is a diagram of the feeding mechanism viewed from the side. FIG. **11** is a perspective view of the feed arm viewed from the upstream side.

As illustrated in FIG. **9**, the feed arm **122** further includes a press roller **122***c* and a second protruding part **122***d*.

As illustrated in FIG. **9** and FIG. **10**, the press roller **122***c* faces the feed roller **112** and is located on the upstream side of the nip part of the feed roller **112** and the separation roller **113** in the medium conveying direction A**1**. The press roller **122***c* is located in the neighborhood of the upstream edge P**1** of the protruding part **122***a* in the medium conveying direction A**1**. The press roller **122***c* presses a medium fed by the feed roller **112** to the feed roller **112** side from above. In the example illustrated in FIG. **9** and FIG. **10**, the feed arm **122** includes two press rollers **122***c*. The press roller **122***c* and the feed roller **112** sandwich the medium in between, and the press roller **122***c* provides a conveyance force to the medium fed by the feed roller **112**. Consequently, the medium conveying apparatus **100** can satisfactorily feed the medium.

The press roller **122***c* is provided on the feed arm **122** on which the protruding part **122***a* is provided and is provided to move in conjunction with the protruding part **122***a*. Consequently, the medium conveying apparatus **100** can simply control the feed arm **122** and the press roller **122***c*, which enables reduction in the development cost of the medium conveying apparatus **100**. Further, by forming the protruding part **122***a* and the press roller **122***c* as one part, the medium conveying apparatus **100** can reduce the apparatus cost and the apparatus weight.

By the central part of a medium being pressed down by the press roller **122***c* in the width direction A**8**, the outer part of the medium is more likely to lift. However, since the outer part of the medium is pressed down by the protruding part **122***a*, the medium conveying apparatus **100** can suppress occurrence of buckling of the medium due to the side of a lifting medium coming in contact with a rubber surface of the separation roller **113**.

As illustrated in FIG. **9** and FIG. **11**, the second protruding part **122***d* faces the feed roller **112** and is located on the upstream side of the nip part of the feed roller **112** and the separation roller **113** in the medium conveying direction A**1**. In other words, the second protruding part **122***d* is positioned on the separation roller **113** side of (above) the nip part of the feed roller **112** and the separation roller **113**. The second protruding part **122***d* guides a medium fed by the feed roller **112** to the nip part of the feed roller **112** and the separation roller **113**. In the example illustrated in FIG. **9** and FIG. **11**, the feed arm **122** includes two second protruding parts **122***d* located to sandwich each press roller **122***c* in between in the width direction A**8** for one feed roller **112** and includes four second protruding parts **122***d* in total.

A fed medium is guided to the nip part of the feed roller **112** and the separation roller **113** by the second protruding part **122***d*. In particular, even when the fed medium is, for example, a medium of which front edge is curled upward or thin paper, the medium conveying apparatus **100** can reliably guide the medium to the nip part. The medium conveying apparatus **100** can press down a lift of a part of a medium not facing the feed roller **112** by the protruding part **122***a* while pressing down a lift of a part of the medium facing the feed roller **112** by the second protruding part **122***d*. Consequently, the medium conveying apparatus **100** can more satisfactorily suppress occurrence of a lift of the front edge of a medium and can suppress occurrence of buckling of a medium with a lifting front edge due to the medium being flipped up by the separation roller **113**.

In particular, the second protruding part **122***d* is located to overlap the press roller **122***c* viewed from the width direction A**8**. In other words, in the medium conveying direction A**1**, the upstream edge of the second protruding part **122***d* is located in the neighborhood of the press roller **122***c*, and the downstream edge of the second protruding part **122***d* is located in the neighborhood of the nip part of the feed roller **112** and the separation roller **113**. Consequently, the second protruding part **122***d* can suitably guide a medium pressed by the press roller **122***c* to the nip part of the feed roller **112** and the separation roller **113**. The upstream edge of the second protruding part **122***d* is located on the downstream side of the press roller **122***c*. Alternatively, the upstream edge of the second protruding part **122***d* may be located on the upstream side of the press roller **122***c*.

The press roller **122***c* and/or the second protruding part **122***d* may be omitted.

FIG. **12** is a schematic diagram for illustrating an example of a setting guide and a flap. FIG. **12** is a diagram of the feeding mechanism before medium feeding viewed from the side.

As illustrated in FIG. **12**, the feeding mechanism **120** further includes the setting guide **124** and the flap **125**.

The setting guide **124** is an example of a second guide member and is a guide for setting a medium. The setting guide **124** is located at a position facing the feed roller **112** and the separation roller **113** in the medium conveying direction A**1**. The setting guide **124** is provided in the lower housing **101** in such a way as to be able to swing (rotate) downward (in a direction of an arrow A**9** in FIG. **12**) according to a driving force from an unillustrated motor. Before medium feeding, the setting guide **124** is located at a first position (a placement position in FIG. **12**) where contact between a medium M**2** placed in the loading tray **103**, and the feed roller **112** and the press roller **122***c* is restricted and supports the bottom surface of the medium M**2** placed in the loading tray **103**.

The flap **125** is a stopper for preventing the medium M**2** from entering a nip part of the feed roller **112** and the separation roller **113** before medium feeding. The flap **125** is located at a position facing the setting guide **124** in the medium conveying direction A**1**. The flap **125** is provided on the feed arm **122** in such a way as to be able to swing (rotate) to the downstream side (in a direction of an arrow A**10** in FIG. **12**) and is pressed toward the upstream side (a direction opposite to the arrow A10) by an unillustrated spring. Before medium feeding, the flap **125** is engaged with the setting guide **124** located at the first position and prevents the medium M2 from entering the nip position of the feed roller **112** and the separation roller **113**. The flap **125** is provided to work with the feed arm **122**; and when the flap **125** is engaged with the setting guide **124**, the feed arm **122** is supported by the flap **125** and the setting guide **124**, and downward movement of the feed arm **122** is prevented. Therefore, the feed arm **122** is stored in the separation roller cover **123** in FIG. **12**

FIG. **13** is a schematic diagram for illustrating the setting guide and the flap during medium feeding. FIG. **13** is a schematic diagram of the feeding mechanism during medium feeding viewed from the side.

As illustrated in FIG. **13**, the setting guide **124** swings below the conveyance surface **121***a* (in the direction of the arrow A**9**) when medium feeding is started. Consequently, at the time of medium feeding, the setting guide **124** is located at a second position (a placement position in FIG. **13**) where contact between the medium M2 placed in the loading tray **103**, and the feed roller **112** and the press roller **122***c* is allowed and separates from the bottom surface of the medium M2 placed in the loading tray **103**.

By the setting guide **124** being located at the second position, the engagement between the flap **125** and the setting guide **124** is released. Consequently, the flap **125** is pushed by the front edge of the medium M2 placed in the loading tray **103** and swings to the downstream side (the direction of the arrow A**10**), and the medium M2 becomes able to enter the nip part of the feed roller **114** and the separation roller **115**. Thus, the flap **125** allows entry of the medium M1 into the nip part of the feed roller **112** and the separation roller **113** when the setting guide **124** is located at the second position.

As described above, since the feed arm **122** is urged downward by the elastic member, the feed arm **122** moves downward (to the feed roller **112** side) by release of the engagement between the flap **125** and the setting guide **124**. In the example illustrated in FIG. **12** and FIG. **13**, an amount of media M2 placed in the loading tray **103** is sufficiently small. In this case, first, the feed roller **112** comes in contact with a medium placed lowest out of the media M2 placed in the loading tray **103**, and the press roller **122***c* subsequently comes in contact with a medium placed uppermost out of the media M2 placed in the loading tray **103**. In other words, the press roller **122***c* is provided to come in contact with a medium placed in the loading tray **103** after the feed roller **112** comes in contact with the medium when an amount of media placed in the loading tray **103** is less than a predetermined amount, at a time when the setting guide **124** moves from the first position to the second position.

When rotation of the feed roller **112** is started in a state of the press roller **122***c* pressing a medium in a case of an amount of media placed in the loading tray **103** being small, the front edge of the medium is likely to be bent upward and thus a jam of the medium is likely to occur. By causing the feed roller **112** to come in contact with a medium before the press roller **122***c* and start feeding of the medium when an amount of media is less than the predetermined amount, the medium conveying apparatus **100** can suppress occurrence of a jam of the medium due to upward bending of the front edge of the medium.

FIG. **14** is a schematic diagram for illustrating the setting guide and the flap when a large amount of media M3 are placed in the loading tray. FIG. **14** is a schematic diagram of the feeding mechanism viewed from the side immediately after medium feeding starts when a large amount of the media M3 are placed in the loading tray.

As described above, when medium feeding starts, the setting guide **124** is moved to the second position, the engagement between the flap **125** and the setting guide **124** is released, and the feed arm **122** moves downward. As illustrated in FIG. **14**, when a large amount of the media M3 are placed in the loading tray **103**, the feed arm **122** is lowered before the setting guide **124** is fully lowered. Therefore, the press roller **122***c* comes in contact with a medium placed uppermost out of the media M**3** placed in the loading tray **103** before the feed roller **112** comes in contact with a medium placed lowest out of the medium M**3** placed in the loading tray **103**. In other words, the press roller **122***c* is provided to come in contact with a medium placed in the loading tray **103** before the feed roller **114** comes in contact with the medium when an amount of media placed in the loading tray **103** is greater than or equal to a predetermined amount, at a time when the setting guide **124** moves from the first position to the second position.

When an amount of media placed in the loading tray **103** is large, the feed roller **114** can satisfactorily send out a medium by starting rotation of the feed roller **114** in a state of the press roller **122***c* pressing the medium. In particular, when an amount of media placed in the loading tray **103** is large, stronger urging force is provided to a medium by the elastic member compared with a case where the amount of media is small. For example, when the elastic member is a compression spring, the magnitude of the urging force is a multiplied value acquired by multiplying a contracted amount of the spring by a spring constant. When an amount of media placed in the loading tray **103** is large, the contracted amount of the spring is large and strong urging force is provided to a medium by the press roller **122***c*, and therefore the feed roller **112** can satisfactorily feed the medium.

When an amount of media placed in the loading tray **103** is large, the front edge of a feed target medium existing at the lowest position is pressed down by the weight of media placed thereon, and therefore the possibility of the front edge of the target medium being bent upward is low. Accordingly, when an amount of media placed in the loading tray **103** is large and the possibility of occurrence of a jam of a medium is low, the medium conveying apparatus **100** can satisfactorily feed a medium while giving priority to feedability of a medium.

FIG. **15** is a block diagram illustrating a schematic configuration of the medium conveying apparatus.

In addition to the configuration described above, the medium conveying apparatus **100** further includes a motor **131**, an interface device **132**, a storage device **140**, a processing circuit **150**, etc.

The motor **131** includes one or a plurality of motors and conveys a medium by rotating the feed roller **112**, the separation roller **113**, the first conveyance roller **114**, the second conveyance roller **115**, the first ejection roller **117**, and the second ejection roller **118** in accordance with a control signal from the processing circuit **150**. One of the first conveyance roller **114** and the second conveyance roller **115** may be a driven roller driven to rotate by the other roller. One of the first ejection roller **117** and the second ejection roller **118** may be a driven roller driven to rotate by the other roller. The motor **131** moves the bottom surface guide **124** between the first position and the second position in accordance with the control signal from the processing circuit **150**.

For example, the interface device **132** includes an interface circuit conforming to a serial bus such as USB and transmits and receives an input image and various types of information by being electrically connected to an unillustrated information processing apparatus (such as a personal computer or a mobile information terminal). A communication device including an antenna transmitting and receiving wireless signals and a wireless communication interface circuit for transmitting and receiving signals through a wireless communication line in accordance with a predetermined communication protocol may be used in place of the interface device **132**. For example, the predetermined communication protocol is a wireless local area network (LAN). The communication device may include a wired communication interface circuit for transmitting and receiving signals through a wired communication line in accordance with a communication protocol such as a wired LAN.

The storage device **140** includes a memory device such as a random-access memory (RAM) or a read-only memory (ROM), a fixed disk device such as a hard disk, a portable storage device such as a flexible disk or an optical disk, etc. Further, a computer program, a database, a table, etc., that are used for various types of processing in the medium conveying apparatus **100** are stored in the storage device **140**. The computer programs may be installed on the storage device **140** from a computer-readable, non-transitory portable storage medium by using a well-known set-up program, etc. The portable storage medium is, for example, a compact disc read-only memory (CD-ROM) or a digital versatile disc read-only memory (DVD-ROM).

The processing circuit **150** operates in accordance with a program previously stored in the storage device **140**. For example, the processing circuit is a central processing unit (CPU). Examples of the processing circuit **150** that may also be used include a digital signal processor (DSP), a large scale integration (LSI), an application specific integrated circuit (ASIC), and a field-programmable gate array (FPGA).

The processing circuit **150** is connected to the operation device **105**, the display device **106**, the medium sensor **111**, the imaging device **116**, the motor **131**, the interface device **132**, the storage device **140**, etc., and controls the components. The processing circuit **150** performs drive control of the motor **131**, imaging control of the imaging device **116**, etc., based on the medium signal received from the medium sensor **111**, acquires an input image from the imaging device **116**, and transmits the acquired image to the information processing apparatus through the interface device **132**.

FIG. **16** is a diagram illustrating a schematic configuration of the storage device land the processing circuit.

As illustrated in FIG. **16**, a control program **141** and an image acquisition program **142**, etc., are stored in the storage device **140**. Each program is a functional module implemented by software operating on the processor. The processing circuit **150** reads each program stored in the storage device **140** and operates in accordance with the read program. Consequently, the processing circuit **150** functions as a control module **151** and an image acquisition module **152**.

FIG. **17** is a flowchart illustrating an operation example of medium reading processing in the medium conveying apparatus.

The operation example of the medium reading process in the medium conveying apparatus **100** will be described below referring to the flowchart illustrated in FIG. **12**. The operation flow described below is executed mainly by the processing circuit **150** in accordance with a program previously stored in the storage device **140** in cooperation with the components in the medium conveying apparatus **100**. The setting guide **124** is located at the first position before execution of the flowchart illustrated in FIG. **12**, i.e., before medium feeding.

First, the control module **151** waits until an instruction to read a medium is input by a user by using the operation device **105** or the information processing apparatus and an operation signal providing an instruction to read a medium is received from the operation device **105** or the interface device **132** (step S**101**).

Next, the control module **151** acquires the medium signal from the medium sensor **113** and determines whether a medium exists on the loading tray **103**, based on the acquired medium signal (step S**102**). When a medium does not exist on the loading tray **103**, the control module **151** ends the series of steps.

On the other hand, when a medium exists on the loading tray **103**, the control module **151** moves the bottom surface guide **124** from the first position to the second position by driving the motor **131**. Further, the control module **151** conveys the medium by rotating the feed roller **112**, the separation roller **113**, the first conveyance roller **114**, the second conveyance roller **115**, the first ejection roller **117**, and/or the second ejection roller **118** by driving the motor **131** (step S**103**).

Next, the control module **151** acquires an input image from the imaging device **116** by causing the imaging device **116** to image the medium and outputs the acquired input image by transmitting the image to the information processing apparatus through the interface device **132** (step S**104**).

Next, the control module **151** determines whether a medium remains on the loading tray **103**, based on the medium signal received from the medium sensor **113** (step S**105**). When a medium remains in the loading tray **103**, the control module **151** returns the processing to step S**104** and repeats the processing in step S**104** and S**105**.

On the other hand, when a medium does not remain on the loading tray **103**, the control module **151** controls the motor **131** to stop the feed roller **112**, the separation roller **113**, the first conveyance roller **114**, the second conveyance roller **115**, the first ejection roller **117**, and/or the second ejection roller **118**. Further, the control module **151** controls the motor **131** to move the setting guide **124** from the second position to the first position (step S**106**) and ends the series of steps.

As described in detail above, the medium conveying apparatus **100** is provided with the recessed part **121***c* in a region around the feed roller **112** in the conveyance surface **121***a* of a medium. Furthermore, the medium conveying apparatus **100** is provided with the protruding part **122***a* guiding a region in the medium outside the feed roller **112** to a region on the feed roller **112** side of the nip part of the feed roller **112** and the separation roller **113**. Consequently, when various types of media, such as thin paper, an envelope, and carbon paper, are conveyed, the medium conveying apparatus **100** can suppress occurrence of jamming of a medium. Accordingly, the medium conveying apparatus **100** can suitably feed a plurality of types of media.

The medium conveying apparatus **100** can convey a bound medium, such as an envelope or carbon paper, even when the feed mode is set to the non-separation mode. Consequently, the medium conveying apparatus **100** can successively convey a plurality of envelopes or a plurality of sheets of carbon paper. A user does not need to change the setting of the feed mode when a bound medium, such as an envelope or carbon paper, is conveyed; and the medium conveying apparatus **100** can improve user convenience.

FIG. **18** is a schematic diagram for illustrating an example of a protruding part in a medium conveying apparatus according to another embodiment FIG. **18** is a schematic diagram of an example of a feeding mechanism viewed from the downstream side. In FIG. **18**, parts other than the protruding part **222***a* of a feed arm is omitted for enhanced visual recognizability.

As illustrated in FIG. **18**, the medium conveying apparatus according to the present embodiment includes the feeding mechanism **220** in place of the feeding mechanism **120**, and the feeding mechanism **220** includes the protruding part **222***a* in place of the protruding part **122***a*. The protruding part **222***a* has a structure similar to that of the protruding part **122***a*.

The tip (lower end) of the protruding part **222***a* has a rounded shape. In other words, a contact surface of the protruding part **222***a* coming in contact with a medium fed by a feed roller **112** is R-shaped. Consequently, when non-carbon paper is fed as a medium M**4**, the medium conveying apparatus can suppress color development due to the non-carbon paper coming in contact with the tip of the protruding part **222***a*. Furthermore, for example, when an envelope or thick paper is fed as a medium, the medium conveying apparatus can suppress appearing a vertical line on the medium by the tip of the protruding part **222***a*.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the R-shaped protruding part **222***a* is used as well.

FIG. **19**A and FIG. **19**B are schematic diagrams for illustrating an example of a protruding part in a medium conveying apparatus according to yet another embodiment FIG. **19**A and FIG. **19**B are schematic diagrams of a feeding mechanism viewed from the downstream side. In FIG. **19**A and FIG. **19**B, parts other than the protruding part **322***a* of a feed arm is omitted for enhanced visual recognizability. FIG. **19**A illustrates a situation in which a medium M**5** with low stiffness, such as thin paper or carbon paper, is fed, and FIG. **19**B illustrates a situation in which a medium M**6** with high stiffness, such as thick paper, is fed.

As illustrated in FIG. **19**A and FIG. **19**B, the medium conveying apparatus according to the present embodiment includes the feeding mechanism **320** in place of the feeding mechanism **120**, and the feeding mechanism **320** includes the protruding part **322***a* in place of the protruding part **122***a*. The protruding part **322***a* has a structure similar to that of the protruding part **122***a*.

The protruding part **322***a* includes a fixed part **322***e* and a movable part **322***f*. For example, the fixed part **322***e* and the movable part **322***f* are formed of a stiff substance, such as metal. The fixed part **322***e* is fixed to the feed arm. The movable part **322***f* is supported by the fixed part **322***e* through an elastic member **322***g* in such a way as to be able to swing outward in a width direction A**8** (in a direction of an arrow A**11**) For example, the elastic member **322***g* is a helical torsion coil spring and provides a force towards the inward side in the width direction A**8** (in a direction opposite to the arrow A**11**) to the movable part **322***f* in such a way that the movable part **322***f* stops at a position almost perpendicular to a conveyance surface **121***a* (a placement position in FIG. **19**A) by an unillustrated stopper.

As illustrated in FIG. **19**A, when stiffness of the fed medium M**5** is low, the medium M**5** is bent by the protruding part **322***a*. Accordingly, the medium conveying apparatus **100** can stiffen the medium M**5** with low stiffness and can improve stiffness of the medium M**5** traveling in a medium conveying direction A**1**. On the other hand, as illustrated in FIG. **19**B, when stiffness of the fed medium M**6** is high, the protruding part **322***a* swings outward (in the direction of the arrow A**11**) in the width direction A**8** by the medium M**6**. Accordingly, the medium conveying apparatus **100** can suppress damage to the medium M**6** due to deformation of the medium M**6** with high stiffness by the protruding part **322***a*. With regard to the medium M**6** with high stiffness, the possibility of occurrence of buckling and jamming of a medium is sufficiently low even without stiffening by the protruding part **322***a*.

Thus, the protruding part **322***a* has elasticity as a whole. Consequently, the medium conveying apparatus can suppress occurrence of damage to a medium with high stiffness while suppressing occurrence of jamming of a medium with low stiffness.

The protruding part **322***a* may be integrally formed of a flexible substance, such as a rubber or a resin. In that case, the protruding part **322***a* has elasticity as one body. In that case, the medium conveying apparatus can suppress occurrence of damage to a medium with high stiffness while suppressing occurrence of jamming of a medium with low stiffness as well. Further, by the protruding part **322***a* being formed as one body, the medium conveying apparatus can reduce the apparatus cost and the apparatus weight.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the protruding part **322***a* with elasticity is used as well.

FIG. **20**A and FIG. **20**B are schematic diagrams for illustrating an example of a guide member in a medium conveying apparatus according to yet another embodiment. FIG. **20**A and FIG. **20**B are schematic diagrams of an example of a feeding mechanism viewed from the downstream side. In FIG. **20**A and FIG. **20**B, parts other than a protruding part **122***a* of a feed arm **122** is omitted for enhanced visual recognizability. FIG. **20**A illustrates a situation in which a light medium M**7**, such as thin paper or carbon paper, is fed, and FIG. **20**B illustrates a situation in which a heavy medium M**8**, such as thick paper, is fed.

As illustrated in FIG. **20**A and FIG. **20**B, the medium conveying apparatus according to the present embodiment includes the feeding mechanism **420** in place of the feeding mechanism **120**, and the feeding mechanism **420** includes the guide member **421** in place of the guide member **121**. The guide member **421** has a structure similar to that of the guide member **121**.

The guide member **421** includes moving parts **421***d* located, in a width direction A**8** perpendicular to a medium conveying direction A**1**, outside recessed parts **421***c* positioned on the left and right sides of a feed roller **112** relative to the medium conveying direction. An elastic member **421***e* is provided between the moving part **421***d* and the base of the guide member **421**. For example, the elastic member **421***e* is a helical compression spring or a rubber and provides an upward force (in a direction of an arrow A**12**) to the moving part **421***d*. The moving part **421***d* pushes back the elastic member **421***e* and moves downward by the weight of a medium existing on the moving part **421***d*. In other words, the moving part **421***d* is provided in such a way as to be movable in a direction perpendicular to a conveyance surface **421***a* according to the weight of the fed medium.

Consequently, when the low-weight medium M**7**, such as thin paper, is fed, the placement position of the moving part **421***d* rises, and the level difference between the top surface of the moving part **421***d* and the recessed part **421***c* sufficiently increases. Therefore, the medium conveying apparatus can sufficiently stiffen the medium M**7**. On the other hand, when the heavy-weight medium M**8**, such as thick paper, is fed, the placement position of the moving part **421***d* lowers, and the level difference between the top surface of the moving part **421***d* and the recessed part **421***c* sufficiently decreases. Therefore, the medium conveying apparatus can suppress occurrence of damage to the medium M**8** due to deformation of the medium M**8** by the recessed part **421***c*.

As illustrated in FIG. **20**A, when the elastic member **421***e* is not compressed, the moving part **421***d* is located at a first position (the placement position illustrated in FIG. **20**A) higher than the top surface of the feed roller **112**. Consequently, when the light-weight medium M**7**, such as thin paper, is fed, the level difference between the top surface of the moving part **421***d* and the recessed part **421***c* sufficiently increases, and therefore the medium conveying apparatus can sufficiently stiffen the medium M**7**.

On the other hand, as illustrated in FIG. **20**B, when the elastic member **421***e* is compressed to the maximum, the moving part **421***d* is located at a second position (the placement position illustrated in FIG. **20**B) lower than the top surface of the feed roller **112**. Consequently, when the heavy-weight medium M**8**, such as thick paper, is fed, the feed roller **112** protrudes relative to the top surface of the moving part **421***d*, and therefore the medium conveying apparatus can provide a sufficient conveyance force to the medium M**8** and satisfactorily feed the medium M**8**.

Thus, the moving part **421***d* is provided in such a way as to be movable between the first position higher than the top surface of the feed roller **112** and the second position lower than the top surface of the feed roller **112** in a direction A**12** perpendicular to the conveyance surface **421***a*.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the guide member **421** is provided in a movable manner as well.

FIG. **21**A is a schematic diagram for illustrating an example of a feed arm in a medium conveying apparatus according to yet another embodiment.

As illustrated in FIG. **21**A, the medium conveying apparatus according to the present embodiment includes a feeding mechanism **520** in place of the feeding mechanism **120**, and the feeding mechanism **520** includes a feed arm **522** in place of the feed arm **122**. The feed arm **522** has a structure similar to that of the feed arm **122**.

The feed arm **522** includes a protruding part **522***a*, a press roller **522***c*, and a base **522***e*. The base **522***e* is provided with fixed to an upper housing **102**. The protruding part **522***a* and the press roller **522***c* have structures similar to those of the protruding part **122***a* and the press roller **122***c* that are included in the feed arm **122**, respectively. The protruding part **522***a* is supported by the base **522***e* through a first elastic member **522***h*. For example, the first elastic member **522***h* is a helical compression spring or a rubber and provides a downward force to the protruding part **522***a*. The press roller **522***c* is supported by the base **522***e* through a second elastic member **522***i*. For example, the second elastic member **522***i* is a helical compression spring or a rubber and provides a downward force to the press roller **522***c*.

Thus, the press roller **522***c* is supported by the base **522***e* through the second elastic member **522***i* separate from the first elastic member **522***h* provided between the base **522***e* and the protruding part **522***a*. Consequently, the press roller **522***c* is provided to move independently of the protruding part **522***a*.

In order to satisfactorily convey a medium with high stiffness, such as thick paper, a conveyance force greater than that for conveying a medium with low stiffness is generally required. In a case of the press roller moving in conjunction with the protruding part, when a medium with high stiffness is conveyed, the protruding part is pushed up by the medium and the press roller consequently rises (lifts), which reduces the force pressing the medium and reduces the conveyance force. On the other hand, even when the protruding part **522***a* is pushed up by a medium with high stiffness in a case of the press roller **522***c* moving independently of the protruding part **522***a*, the press roller **522***c* does not rise and continues to press the medium. Accordingly, the medium conveying apparatus can satisfactorily convey a medium with high stiffness, such as thick paper, while suppressing occurrence of jamming of a medium with low stiffness, such as thin paper.

As illustrated in FIG. **21**A, the protruding part **522***a* may be provided on the downstream side of the press roller **522***c* in a medium conveying direction A**1**. In a case of the press roller moving in conjunction with the protruding part, when a thin medium is conveyed subsequently to conveyance of a thick medium, such as thick paper, the protruding part is pushed up by the rear edge of the preceding medium, and the press roller may consequently rise and may separate from the front edge of the succeeding medium. On the other hand, even when the protruding part **522***a* is pushed up by the preceding medium in a case of the press roller **522***c* moving independently of the protruding part **522***a*, the press roller **522***c* does not rise and continues to press the succeeding medium. Accordingly, even when a plurality of media with different thicknesses are successively fed, the medium conveying apparatus can satisfactorily feed each medium.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the press roller **522***c* is provided to move independently of the protruding part **522***a* as well.

FIG. **21**B is a schematic diagram for illustrating an example of a feed arm **622** in a medium conveying apparatus according to yet another embodiment.

As illustrated in FIG. **21**B, the medium conveying apparatus according to the present embodiment includes a feeding mechanism **620** in place of the feeding mechanism **520**, and the feeding mechanism **620** includes the feed arm **622** in place of the feed arm **522**. The feed arm **622** has a structure similar to that of the feed arm **522**.

The feed arm **622** includes a protruding part **622***a*, a press roller **622***c*, and a base **622***e*. The base **622***e* is provided with fixed to an upper housing **102**. The protruding part **622***a*, the press roller **622***c*, and the base **622***e* have structures similar to those of the protruding part **522***a*, the press roller **522***c*, and the base **522***e* that are included in the feed arm **522**, respectively. The protruding part **622***a* is supported by the base **622***e* through a first elastic member **622***h*. For example, the first elastic member **622***h* is a helical compression spring or a rubber and provides a downward force to the protruding part **622***a*. The press roller **622***c* is supported by a member including the protruding part **622***a* through a second elastic member **622***i*. For example, the second elastic member **622***i* is a helical compression spring or a rubber and provides a downward force to the press roller **622***c*.

The press roller **622***c* is provided to move independently of the protruding part **622***a* in this case as well. Accordingly, the medium conveying apparatus can satisfactorily convey a medium with high stiffness, such as thick paper, while suppressing occurrence of jamming of a medium with low stiffness, such as thin paper. Further, even when a plurality of media with different thicknesses are successively fed, the medium conveying apparatus can satisfactorily convey each medium.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the press roller **622***c* is provided to move independently of the protruding part **622***a* as well.

The feed arm **522** or the feed arm **622** may further include a second protruding part similar to the second protruding part **122***d*. In that case, the second protruding part is provided to move in conjunction with the protruding part **522***a* or the protruding part **622***a* and move independently of the press roller **522***c* or the press roller **622***c*.

FIG. **22** is a schematic diagram for illustrating an example of a guide member in a medium conveying apparatus according to yet another embodiment.

As illustrated in FIG. **22**, the medium conveying apparatus according to the present embodiment includes a feeding mechanism **720** in place of the feeding mechanism **120**, and the feeding mechanism **720** includes the guide member **721** in place of the guide member **121**. The guide member **721** has a structure similar to that of the guide member **121**. The guide member **721** includes an opening **721***b* similar to the opening **121***b*, and a conveyance surface **721***a* of the guide member **721** includes a recessed part **721***c* similar to the recessed part **121***c*.

The opening **721***b* is smaller than the opening **121***b*, and the recessed part **721***c* exists up to the neighborhood of a feed roller **112** in a width direction A**8**. Therefore, the protruding part **122***a* is located at a position facing the recessed part **721***c*.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the protruding part **122***a* is located at a position facing the recessed part **721***c* as well.

FIG. **23** is a schematic diagram for illustrating an example of a guide member in a medium conveying apparatus according to yet another embodiment.

As illustrated in FIG. **23**, the medium conveying apparatus according to the present embodiment includes a feeding mechanism **820** in place of the feeding mechanism **120**, and the feeding mechanism **820** includes the guide member **821** in place of the guide member **121**. The guide member **821** has a structure similar to that of the guide member **121**. A conveyance surface **821***a* of the guide member **821** has a recessed part **821***c* similar to the recessed part **121***c*.

The guide member **821** does not include the opening **121***b* and instead includes a notch **821***b*. A feed roller **112** is located inside the notch **821***b*, and the protruding part **122***a* is located at a position facing the notch **821***b*.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the guide member **821** includes the notch **821***b* as well.

FIG. **24** is a schematic diagram for illustrating an example of a guide member **921** in a medium conveying apparatus according to yet another embodiment.

As illustrated in FIG. **24**, the medium conveying apparatus according to the present embodiment includes a feeding mechanism **920** in place of the feeding mechanism **820**, and the feeding mechanism **920** includes the guide member **921** in place of the guide member **821**. The guide member **921** has a structure similar to that of the guide member **821**. The guide member **921** includes a notch **921***b* similar to the notch **821***b*, and a conveyance surface **921***a* of the guide member **921** includes a recessed part **921***c* similar to the recessed part **821***c*.

The notch **921***b* is smaller than the notch **821**, and the recessed part **921***c* exists up to the neighborhood of a feed roller **112** in a width direction A**8**. Therefore, the protruding part **122***a* is located at a position facing the recessed part **921***c*.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the guide member **921** includes the notch **921***b* and the protruding part **122***a* is located at a position facing the recessed part **921***c* as well.

FIG. **25** is a diagram illustrating a schematic configuration of an example of a processing circuit in a medium conveying apparatus according to yet another embodiment. The processing circuit **1050** is used in place of the processing circuit **150** in the medium conveying apparatus **100** and executes the medium reading processing, etc., in place of the processing circuit **150**. The processing circuit **1050** includes a control circuit **1051** and an image acquisition circuit **1052**, etc. Each component may be independently configured with an integrated circuit, a microprocessor, firmware, etc.

The control circuit **1051** is an example of a control module and has a function similar to that of the control module **151**. The control circuit **1051** receives the operation signal from an operation device **105** or an interface device **132**, the medium signal from the medium sensor **111**. The control circuit **1051** controls a motor **131**, based on the received information.

The image acquisition circuit **1052** is an example of an image acquisition module and has a function similar to that of the image acquisition module **152**. The image acquisition circuit **1052** acquires the input image from an imaging device **116** and outputs the image to the interface device **132**.

As described in detail above, the medium conveying apparatus can suitably feed a plurality of types of media when the processing circuit **1050** is used as well.

REFERENCE SIGNS LIST

100 Medium conveying apparatus, **103** Loading tray, **112** Feed roller, **113** Separation roller, **121**, **421** Guide member, **121***a* Conveyance surface, **121***b* Opening, **121***c*, **421***c* Recessed part, **121***b*, **721***b* Opening, **122**, **522**, **622** Feed arm, **122***a*, **222***a*, **322***a*, **522***a*, **622***a* Protruding part, **122***b* Contact surface, **122***c*, **522***c*, **622***c* Press roller, **122***d* Second protruding part, **124** Setting guide, **421***d* Moving part, **821***b*, **921***b* Notch

The invention claimed is:

1. A medium conveying apparatus comprising:

a feed roller to feed a medium;

a separation roller located to face the feed roller and to form a nip part with the feed roller for separating the medium;

a press roller facing the feed roller and located on an upstream side of the nip part formed by the feed roller and the separation roller in a medium conveying direction to press the medium fed by the feed roller toward the feed roller;

protruding parts located on left and right sides of the separation roller relative to the medium conveying direction, and configured to move in conjunction with the press roller, wherein

a portion of at least one of the protruding parts is positioned on the feed roller side of the nip part, and

when viewed from a direction perpendicular to the medium conveying direction, at least a portion of at least one of the protruding parts continuously overlaps with the nip part throughout feeding of the medium.

2. The medium conveying apparatus according to claim **1**, further comprising:

a loading tray; and

a guide member, which is located at a first position where contact between the medium placed in the loading tray, and the feed roller and the press roller is restricted before medium feeding, and is located at a second position where contact between the medium placed in the loading tray, and the feed roller and the press roller is allowed at a time of medium feeding, wherein

the press roller is configured to come in contact with the medium placed in the loading tray before the feed roller comes in contact with the medium when an amount of medium placed in the loading tray is greater than or equal to a predetermined amount, and come in contact with the medium placed in the loading tray after the feed roller comes in contact with the medium when the amount of medium placed in the loading tray is less than the predetermined amount, at a time when the guide member moves from the first position to the second position.

3. The medium conveying apparatus according to claim **1**, further comprising second protruding parts facing the feed roller, located on an upstream side of the nip part in the medium conveying direction, and configured to guide the medium fed by the feed roller to the nip part.

4. The medium conveying apparatus according to claim **1**, further comprising a guide member including an opening, notch, or a recessed part and configured to form a conveyance surface of the medium, wherein a contact surface of at least one of the protruding parts coming in contact with the medium fed by the feed roller is formed in such a way as to be in parallel with a region of the conveyance surface on an upstream side of the opening, the notch, or the recessed part in the medium conveying direction.

5. The medium conveying apparatus according to claim **1**, wherein a contact surface of at least one of the protruding parts coming in contact with the medium fed by the feed roller is R-shaped.

6. The medium conveying apparatus according to claim **1**, wherein the protruding parts have elasticity.

7. The medium conveying apparatus according to claim **1**, further comprising a guide member including an opening, notch, or a recessed part and configured to form a conveyance surface of the medium, wherein the guide member further includes a moving part located outside the opening, the notch, or the recessed part in a direction perpendicular to the medium conveying direction and provided in such a way as to be movable in a direction perpendicular to the conveyance surface according to a weight of the fed medium.

8. The medium conveying apparatus according to claim **7**, wherein the moving part is provided in such a way as to be movable between a first position higher than a top surface of the feed roller and a second position lower than the top surface of the feed roller in a direction perpendicular to the conveyance surface.

9. The medium conveying apparatus according to claim **1**, further comprising a guide member including an opening, notch, or a recessed part and configured to form a conveyance surface of the medium, wherein the protruding parts are located at the positions facing the opening or the notch, or the recessed part in such a way that the medium is conveyed between the protruding parts and the opening or the notch, or the recessed part.

10. The medium conveying apparatus according to claim **1**, further comprising:

a flap;

a guide member movable between a first position and a second position; and

a loading tray for placing the medium to be fed by the feed roller; wherein

the guide member restricts contact between the medium placed on the loading tray and the feed roller when located at the first position, and allows contact between the medium placed in the loading tray and the feed roller when located at the second position, and

the flap is engaged with the guide member when the guide member is located at the first position to prevent the medium from entering the nip part.

11. The medium conveying apparatus according to claim **1**, wherein the protruding parts are not swingable.

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